

THROMBOSIS AND HEMOSTASIS

Targeting ACKR3/CXCR7 enhances platelet anticoagulant acylcarnitines and modulates procoagulant function

Xiaoqing Fu,^{1,2} Adrian Brun,¹ Kristina Dittrich,¹ Malgorzata Cebo,¹ Marcel Lackner,³ Lena-Sophie Menig-Benzig,^{3,4} Benjamin Bouzabia,⁵ Johannes Rheinlaender,⁶ Hadra Banks,⁶ Hendrik von Eysmond,⁶ Tanja Dötsch,⁷ Sandra Schwegmann,⁵ Josue Alan Bucio-Garcia,⁵ Bernd Nürnberg,⁵ Sandra Beer-Hammer,⁵ Matthias Schwab,^{3,8,9} Elke Schaeffeler,³ Ute Hofmann,³ Mathias Haag,³ Dominik Rath,¹⁰ Tobias Geisler,¹⁰ Meinrad Gawaz,¹⁰ Tamam Bakchoul,⁷ Tilman E. Schäffer,⁶ Michael Lämmerhofer,¹ and Madhumita Chatterjee⁷

¹Institute of Pharmaceutical Sciences, University of Tübingen, Tübingen, Germany; ²School of Medicine, Shanghai University, Shanghai, China; ³Dr. Margarete Fischer-Bosch Institute of Clinical Pharmacology, Stuttgart, Germany; ⁴Institute of Medical Genetics and Applied Genomics, University of Tübingen, Tübingen, Germany; ⁵Department of Pharmacology, Experimental Therapy and Toxicology, University Hospital Tübingen, Tübingen, Germany; ⁶Institute of Applied Physics, University of Tübingen, Tübingen, Germany; ⁷Institute for Clinical and Experimental Transfusion Medicine, University Hospital Tübingen, Tübingen, Germany; ⁸Department of Clinical Pharmacology and ⁹Department of Pharmacy and Biochemistry, University of Tübingen, Tübingen, Germany; and ¹⁰Department of Cardiology and Angiology, University Hospital Tübingen, Tübingen, Germany

KEY POINTS

- Therapeutic targeting of CXCR7 regulates intraplatelet nonenzymatic lipid (per)oxidation and preserves mitochondrial metabolism.
- CXCR7 agonism induces anticoagulant CARs and regulates procoagulant platelet functions, thereby reducing venous thrombosis.

Targeting ACKR3/CXCR7 regulates enzymatic generation of prothrombotic lipids while favoring antithrombotic lipids that inhibit platelets through the AC-cAMP-PKA pathway in coordination with prostacyclin IP receptor. This investigation validated the effect of CXCR7 in modulating nonenzymatic lipid (per)oxidation, platelet response to lipoproteins, mitochondrial metabolism, and procoagulant functions. CXCR7 agonist VUF11207 preserved mitochondrial membrane integrity, counteracted activation-induced mitochondrial superoxide generation, and reduced nonenzymatic lipid (per)oxidation. Moreover, it regulated lipoprotein-induced platelet adhesion to thrombogenic matrices, degranulation, $\alpha_{IIb}\beta_{III}$ -integrin activation, aggregation, and thrombotic responses by reducing lipoprotein uptake through CD36 and ApoER2. CXCR7 ligation triggered the activation of AMP-dependent kinase^{Ser-172} and prompted AMPK-mediated inhibitory phosphorylation of acetyl-coenzyme A carboxylase^{Ser-79} to foster lipolysis over lipogenesis. Consequently, the AMPK^{Ser-172}-ACC^{Ser-79} pathway increased generation of anticoagulant FXa-inhibitory long-chain acylcarnitines (LC-CAR) in platelets of healthy subjects and patients with coronary artery disease. Enrichment of intraplatelet LC-CARs was not attributable to dysregulated mitochondrial respiration because VUF11207

improved maximal respiration, spare respiratory capacity, and ATP-linked respiration in thrombin-activated platelets, suggesting sustained mitochondrial metabolism. Exerting a 2-pronged effect on procoagulant function, VUF11207 downregulated phosphatidylserine exposure on activated platelets and reduced FX/FXa binding, while platelet-derived anticoagulant LC-CARs regulated thrombin generation. VUF11207 administration reduced thrombus formation, platelet degranulation, $\alpha_{IIb}\beta_{III}$ -integrin activation, procoagulant activity, and circulating platelet-leukocyte aggregates in murine venous thrombosis model, also decreased plasma procoagulant lipids derived from platelet cyclooxygenase-1 and 12-lipoxygenase (LOX), and leukocyte 5/15-LOX, decreased thromboinflammatory mediators (IL-1 β , IL-6, IFN- γ , TNF- α , and MCP-1), and increased plasma LC-CAR levels. Therefore, pharmacological targeting of CXCR7 could regulate (non)enzymatic lipid processing and promote anticoagulant LC-CAR generation to limit platelet-driven thrombotic propensity and hypercoagulability, also replenish reduced levels of circulatory LC-CARs in patients with STEMI and VTE.

Introduction

Lipids serve as an energy reserve, act as extracellular cues that trigger platelet activation (thromboxane A₂ [TxA₂]) or inhibition (prostacyclin [prostaglandin I₂]), function as signaling intermediates, and instigate procoagulant response.¹⁻³ Patients with coronary artery disease (CAD) exhibit increased thrombotic propensity attributed to hyperactive and procoagulant platelets⁴ that expose thrombogenic phospholipids (PLs), phosphatidylserine (PS), phosphatidylethanolamine, and phosphatidylthreonine.⁵⁻⁸ Besides this, aminophospholipid-enriched, platelet-derived extracellular vesicles in the plasma of patients with CAD and acute coronary syndrome (ACS) support prothrombinase activity.⁵ Like CAD, acutely aggravated procoagulant functions in immunothrombotic complications such as viral infections (eg, COVID-19),⁹ sepsis,¹⁰ and vaccine-induced immune thrombotic thrombocytopenia (VITT)¹¹ are not remedied by aspirin,⁹ which targets only the enzymatic generation of TxA₂ through cyclooxygenase-1 (COX-1). Esterification of 12-lipoxygenase (12-LOX)-generated 12-hydroxyeicosatetraenoic acid (12-HETE) to lysophospholipids also supports procoagulant functions.¹² HETE-esterified PLs (HETE-PLs), a class of enzymatically oxidized phospholipids (eoxPLs), are substantially increased in patients with anti-phospholipid syndrome and exhibit immunoreactivity with anti-phospholipid IgG antibodies (β₂-glycoprotein 1).¹² eoxPLs are substantially influenced by aspirin, but not P₂Y₁₂ antagonists.¹³ Platelets from aspirin-treated patients with atherosclerotic cardiovascular disease have lower COX-derived eoxPLs but increased 12-LOX-generated eoxPLs.¹³ Limb and coronary thrombi predominantly exhibit platelet-specific eoxPL (12-HETE-PL) deposition, whereas carotid thrombi show a prevalence of leukocyte-derived eoxPLs (15-, 11-, 5-HETE-PLs).¹³ These clinical insights reveal the insufficiency of current antiplatelet therapies in regulating procoagulant platelets, warranting novel therapeutics. Previously, we demonstrated that targeting ACKR3/CXCR7 with the specific, high-affinity (pK_i = 8.1) pharmacological agonist VUF11207, which does not influence the surface expression of CXCR1, CXCR2, CXCR4, CXCR6, or other functional platelet receptors (GPIb, GPV, GPIX, GPVI, α_v-integrin, β₃-integrin), regulates arterial thrombosis and thromboinflammatory platelet functions.^{14,15} The antithrombotic effects of VUF11207 were absent in megakaryocyte/platelet-specific *Ackr3*-deficient (*PF4-Cre*⁺; *Ackr3*^{fl/fl}) mice, confirming its mode of action primarily through platelet CXCR7. Although CXCR7 agonist does not compromise plasma coagulation or hemostasis, it restrains PS externalization and procoagulant platelet-assisted thrombin generation.¹⁴ Current investigation revealed an unexpected influence of CXCR7 in inducing anticoagulant long-chain acylcarnitine (LC-CAR)¹⁶ in platelets, circulating levels of which are significantly reduced in patients with venous thromboembolism (VTE),¹⁶ ACS (ST-segment elevation myocardial infarction [STEMI]),¹⁷ and VITT.¹⁸

Platelets process lipids through both enzymatic and nonenzymatic means, assisted by reactive oxygen species (ROS) and redox imbalance, the latter remaining unaffected by drugs targeting enzymatic pathways through COX-1, such as aspirin, or 12-LOX, such as ML355.¹⁹ Further, platelet-lipoprotein interaction influences their metabolic status^{3,20} and lowers their activation threshold.²¹⁻²⁴ Platelet scavenger receptors (SRs) engage with circulating lipids.^{21,22} Enhanced CD36

expression mediates increased platelet reactivity to oxidized low-density lipoprotein (oxLDL) and, consequently, thrombotic propensity.²⁵⁻²⁷ In patients with ACS, oxLDL binding correlates significantly with platelet activation status.²⁸ Platelet interaction with plasma lipoproteins also generates ROS,²⁹ facilitating nonenzymatic lipid (per)oxidation and producing oxidized PLs (oxPLs) in patients with CAD.²⁹ Increased levels of phosphatidylcholine 18:0 (10:0/8:0) are associated with advanced disease severity in patients with ACS because they trigger platelet activation,³⁰ whereas platelets from patients with CAD show increased levels of nonenzymatically derived oxPLs with adverse cardiovascular effects.²⁹ Intraplatelet oxLDL levels are enhanced in patients with ACS and are prominently deposited at intracoronary thrombi.²⁹ Therefore, both enzymatic and ROS-mediated nonenzymatic lipid (per)oxidation should be targeted by innovative therapeutics, currently beyond the influence of lipid-lowering and antiplatelet strategies.

Platelet CXCR4-CXCR7 surface expression is enhanced in patients with CAD who have higher plasma low-density lipoprotein (LDL).¹⁴ Contrary to prothrombotic CXCR4, targeting CXCR7 counteracts thrombotic response.^{14,31,32} Moreover, higher platelet CXCR7 expression is associated with reduced aggregation and improved prognosis in patients with CAD.^{14,15,32} CXCR7 agonist VUF11207 favors generation of the antiplatelet lipid mediator 12-LOX-metabolized 12-hydroxyeicosatrienoic acid (12-HETrE) while restricting that of prothrombotic lipid mediators TxA₂, 12-HETE, and the phospholipase-derived atherogenic lysophosphatidylcholine (LPC).¹⁴ The current investigation expanded these translational findings, demonstrating that CXCR7 agonist regulated nonenzymatic lipid processing, mitochondrial metabolism, procoagulant function, and the thrombotic consequences of platelet-lipoprotein interaction. Moreover, we delineated the influence of CXCR7 agonist on the adenosine 5'-monophosphate (AMP) dependent kinase^{Ser-172} acetyl-coenzyme A carboxylase^{Ser-79} (ACC^{Ser-79}) metabolic axis, which favored lipolysis over lipogenesis, promoted anticoagulant LC-CAR generation, and validated its therapeutic potential in venous thrombosis.

Methods

Details are provided in the supplemental Methods, available on the *Blood* website. Platelet degranulation (CD62P), α_{IIb}β₃-integrin activation (PAC-1/JON-A; flow cytometry),¹⁴ aggregation (impedance aggregometry),¹⁴ thrombus formation (T-TAS),¹⁴ surface expression of CXCR7 and SRs (flow cytometry),²⁹ mitochondrial membrane potential (Δψ_m; tetramethylrhodamine ethyl ester assay),^{29,33} mitochondrial superoxide generation (MitoSOX Red),²⁹ lipid peroxidation (C11-BODIPY^{581/591}),²⁹ Dil (1,1'-dioctadecyl-3,3',3'-tetramethylindocarbocyanine perchlorate)-LDL and Dil-oxLDL uptake,²⁹ intraplatelet oxLDL, annexin V and lactadherin binding, and FX/FXa binding to procoagulant platelets (flow cytometry), thrombin generation, were assessed in whole blood, PRP, or platelet-poor plasma (PPP), as appropriate.¹⁴ Scanning ion conductance microscopy,³⁴ metabolomics, lipidomics, mitochondrial respiration analyses (Mito Fuel and Mito Stress tests),³⁵ immunofluorescence microscopy, and immunoblot analyses were performed using isolated platelets.

Release of thrombotic mediators into murine plasma and from human umbilical vein endothelial cell (HUVEC) into cell supernatants were ascertained by cytometric bead array.

CAD clinical cohort

Patients were enrolled in a consecutive cohort for platelet lipidomics analyses^{14,30} (Table 1). Blood samples were collected from the femoral arterial access site during coronary angiography after participants provided informed consent.^{14,15,30,32} The study was approved by the institutional ethics committee (approval numbers 270/2011BO1 and 237/2018BO2) and was conducted in accordance with the Declaration of Helsinki and Good Clinical Practice guidelines.

Venous thrombosis

C57BL/6J mice were administered with the CXCR7 agonist VUF11207 (100 µg per mouse, IV) or vehicle control (1% dimethyl sulfoxide)¹⁴ before ligation (stasis) of the inferior vena cava (IVC). Functional assessment of platelets and analyses of plasma lipids and thromboinflammatory mediators were conducted after 48 hours. Animal experiments were performed in accordance with German animal welfare legislation, followed the ARRIVE guidelines, and were approved by local authorities (approval no. PH08/18G).

Lipidomics analysis (ultrahigh-performance liquid chromatography–quadrupole time-of-flight tandem mass spectrometry)

Platelets from patients with CAD (n = 107) were processed for untargeted lipidomics analysis.³⁰ The effect of CXCR7 agonist on the lipidome of resting and thrombin-activated platelet was assessed in patients with CAD (n = 15) and healthy controls (n = 25).^{14,30,36,37}

Metabolomics analysis (liquid chromatography–tandem mass spectrometry and hydrophilic interaction liquid chromatography–quadrupole time-of-flight mass spectrometry)

Resting and thrombin-activated platelets from healthy donors (n = 6) were analyzed in the presence or absence of CXCR7 agonist treatment.^{38–43}

Results

CXCR7 prevents nonenzymatic lipid oxidation and peroxidation

Activated platelets generate nonenzymatically derived oxPLs^{3,29} despite statin and aspirin therapy in patients with CAD and, increasingly, in patients with STEMI compared to stable angina pectoris.²⁹ eoxPLs also remain unaffected by

Table 1. Baseline characteristics of patients with CAD enrolled for the analysis of platelet lipidome

Characteristics (n = 122)	ACS (n = 28)	CCS (n = 94)	P value
Age, mean ± SD, y	71.7 ± 12.3	70.6 ± 10.3	.645
Male sex, n (%)	21 (75.0)	69 (73.4)	.866
LVEF at admission, mean ± SD, %	47.5 ± 12.0	50.7 ± 11.8	.269
Cardiovascular risk factors, n (%)			
Arterial hypertension	23 (82.1)	76 (80.9)	.878
HL	11 (39.3)	50 (53.2)	.147
Diabetes mellitus type 2	10 (35.7)	27 (28.7)	.490
Smoking	4 (14.3)	19 (20.2)	.485
Medication on admission, n (%)			
ASA	11 (39.3)	48 (51.1)	.274
Clopidogrel	1 (3.6)	14 (14.9)	.106
Prasugrel	0 (0.0)	4 (4.3)	.264
Ticagrelor	1 (3.6)	11 (11.7)	.200
ACE inhibitors	13 (46.4)	48 (51.1)	.631
ARBs	7 (25.0)	24 (25.5)	.932
Beta blockers	17 (60.7)	62 (66.0)	.562
Statins	14 (50.0)	65 (69.1)	.053
Plasma lipid profile, mean ± SD, mg/dL			
Total cholesterol	156.5 ± 53.0	164.5 ± 46.4	.504
LDL cholesterol	97.8 ± 32.9	99.0 ± 38.9	.892
HDL cholesterol	45.7 ± 16.0	49.1 ± 17.2	.429
Triglycerides	114.2 ± 32.4	152.1 ± 92.6	.003

Bold denotes that the comparison is significantly different according to the P value.

ACE, angiotensin-converting enzyme; ARB, angiotensin receptor blocker; ASA, acetylsalicylic acid; HDL, high-density lipoprotein; LVEF, lower left ventricular ejection fraction; SD, standard deviation.

disease severity (chronic coronary syndrome [CCS] vs ACS), hyperlipidemia (HL), or diabetes mellitus but are unexpectedly elevated by antiplatelet therapy (supplemental Figure 1). Prothrombotic CXCL12/SDF1- α synergistically elevates LDL- or oxLDL-induced ROS generation, enabling intraplatelet LDL oxidation.²⁹ However, the pharmacological CXCR7 agonist VUF11207 reduced intraplatelet oxidation of LDL to oxLDL, as assessed by anti-oxLDL fluorescein isothiocyanate staining, after it was induced by LDL and collagen-related peptides (CRP) (Figure 1A), like iloprost, which triggers the prostacyclin receptor (IP receptor)–adenylyl cyclase (AC)–cyclic adenosine monophosphate (cAMP)–protein kinase A (PKA) pathway, and the cell-permeable superoxide dismutase mimetic MnTMPyP (manganese(III) tetrakis(1-methyl-4-pyridyl)porphyrin). Similarly, lipid peroxidation induced by CRP or lipoproteins (LDL or oxLDL), assessed using the lipid peroxidation sensor C11-BODIPY^{581/591}, was also reduced by CXCR7 agonist (Figure 1Bi). Lipoprotein-treated fibrinogen-adherent platelets showed a gradual loss of C11-BODIPY^{581/591} fluorescence, indicating lipid peroxidation, which was significantly counteracted by CXCR7 agonist (Figure 1Bii; supplemental Figure 2A). Moreover, lipoprotein-induced mitochondrial $\Delta\psi_m$ and superoxide generation were significantly reduced by the CXCR7 agonist VUF11207, iloprost, and MnTMPyP (Figure 1C–D). Regulation of nonenzymatic lipid (per)oxidation added to the inhibitory effect of CXCR7 agonist on enzymatically derived thromboinflammatory lipids.¹⁴ Therefore, enhanced surface availability of platelet CXCR7 in patients with CAD having increased plasma LDL levels¹⁴ offers the possibility of regulating nonenzymatic lipid processing induced by circulating lipoproteins.

CXCR7 modulates thrombotic response to lipoproteins

Contrary to CXCL12/SDF1- α , which synergistically enhances lipoprotein-induced platelet activation and arterial thrombosis,²⁹ the priming effects of lipoproteins⁴⁴ on platelet adhesion over collagen- and fibrinogen-coated surfaces (Figure 1E), lipoproteins-induced platelet degranulation (CD62P), $\alpha_{IIb}\beta_{III}$ -integrin activation (PAC-1) (Figure 1Fi), and platelet aggregation²⁹ (Figure 1Fii) were significantly reduced by the CXCR7 agonist VUF11207. CXCR7 agonist also constrained the prothrombotic effects of LDL and oxLDL on collagen-coated surfaces (Figure 1G; supplemental Video 1A–B) and surfaces coated with collagen plus tissue factor (TF) (Figure 1H; supplemental Video 2A–B), reducing thrombus coverage (area under the curve) and prolonging time to occlusion (Figure 1G–H). Thus, CXCR7 agonist may regulate the enhanced thrombotic propensity induced by hyperlipidemia.

CXCR7 agonist regulates platelet-lipoprotein interaction

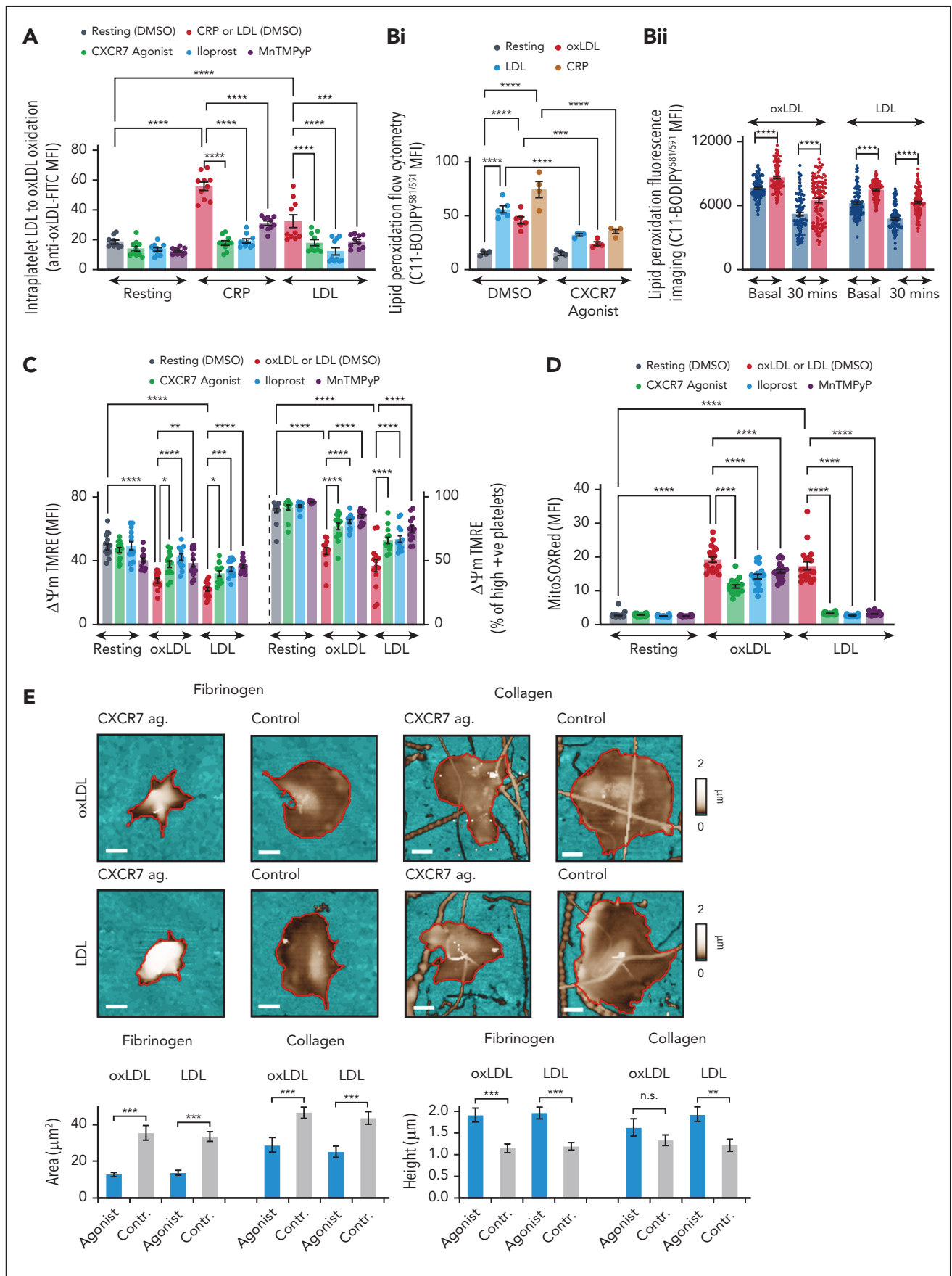
Platelet interaction with plasma lipoproteins is mediated through SRs. CXCL12/SDF1- α facilitates platelet lipoprotein uptake by increasing the surface expression of ApoER2 and CXCL16/SR-PSOX.²⁹ By comparison, administration of the CXCR7 agonist CCX771 in hyperlipidemic (*apoE*^{−/−}) mice promotes very-low-density lipoprotein uptake in white adipose tissue, thereby lowering plasma cholesterol and retarding atheroprogession.⁴⁵ However, contrary to the effects of the prothrombotic inflammatory chemokine CXCL12/SDF1- α , uptake of Dil-LDL or

Dil-oxLDL by platelets was significantly reduced by the CXCR7 agonist VUF11207 (supplemental Figure 2Bi–Bii), attributable to significantly decreased surface expression of ApoER2 and CD36 (supplemental Figure 2C–D), exemplifying a beneficial but cell-specific influence of CXCR7 on lipid uptake. These results also highlight the difference between the platelet-activating signaling cascade triggered by CXCL12/SDF1- α through engagement of G protein α_i -coupled CXCR4,^{46,47} as opposed to the platelet-inhibitory signaling induced by the pharmacological CXCR7-specific agonist VUF11207, acting through the IP receptor–G protein α_s (G α_s)–AC–cAMP–PKA pathway after 12-HETrE binding to the IP receptor.¹⁴

As a fatty acid translocase (FAT), CD36 mediates long-chain fatty acid (LCFA) uptake in diverse cells.^{48–50} The presence of LCFAs in the extracellular milieu/supernatant of CXCR7 agonist-treated platelets could promote their uptake through FAT/CD36, driving CD36 internalization, as indicated by reduced levels of LCFA in supernatant and their corresponding increase in platelets (supplemental Figure 3A). CD36 internalization operates through diverse pathways, varying considerably between ligands and cells.^{48,51–53} Subcellular fractionation studies show CD36 is confined to cholesterol-enriched plasma membrane lipid rafts and that FAT/CD36-mediated LCFA uptake in adipocytes is raft dependent and strongly affected by disruption of lipid rafts with cholesterol-depleting agent methyl- β -cyclodextrin.^{52,53} Our results revealed a requirement for cholesterol-enriched lipid rafts in CXCR7 agonist-mediated CD36 internalization, which was counteracted by methyl- β -cyclodextrin (supplemental Figure 3B). However, methyl- β -cyclodextrin neither inhibited β -arrestin-dependent CXCR7 internalization caused by CXCR7 agonist (supplemental Figure 3C) nor counteracted CXCR7 agonist-mediated regulation of procoagulant (annexin V⁺CD62P⁺) platelet transformation (supplemental Figure 3D). CXCR7 agonist-induced ApoER2 internalization was driven through clathrin-dependent endocytosis^{54,55} and was inhibited by PitStop-2 (supplemental Figure 3E).

CXCR7 agonist induces accumulation of anticoagulant LC-CAR

CXCR7 agonist changes the lipidome of basal and thrombin-activated platelets, favoring generation of 12-LOX-derived antiplatelet lipid 12-HETrE, while restricting prothrombotic 12-HETE and COX-1-metabolized TxA₂.¹⁴ Given the stabilizing effect of CXCR7 agonist on $\Delta\psi_m$, we focused on energy-storage lipids, including triglycerides (TGs), cholesteryl esters (CEs), also diacylglycerols (DGs), and CARs (Figure 2), used through mitochondrial metabolism. Resting and thrombin-activated platelets showed distinct clustering according to their lipid profile (Figure 2Ai), confirming the influence of CXCR7 agonist on the lipidomes of basal/resting and thrombin-activated platelets (Figure 2Aii–iii). TGs were depleted in CXCR7 agonist-treated resting platelets (Figure 2Bi), which also exhibited reduced DG (Figure 2Bii) and CE (Figure 2Biii) levels. Depletion of these lipid reserves was accompanied by a concomitant and significant increase in LC-CAR levels in CXCR7 agonist-treated resting platelets (Figure 2Biv), indicating lipolysis and the consequent conversion of LCFA into LC-CAR by the mitochondrial enzyme carnitine palmitoyl transferase 1 (CPT1).



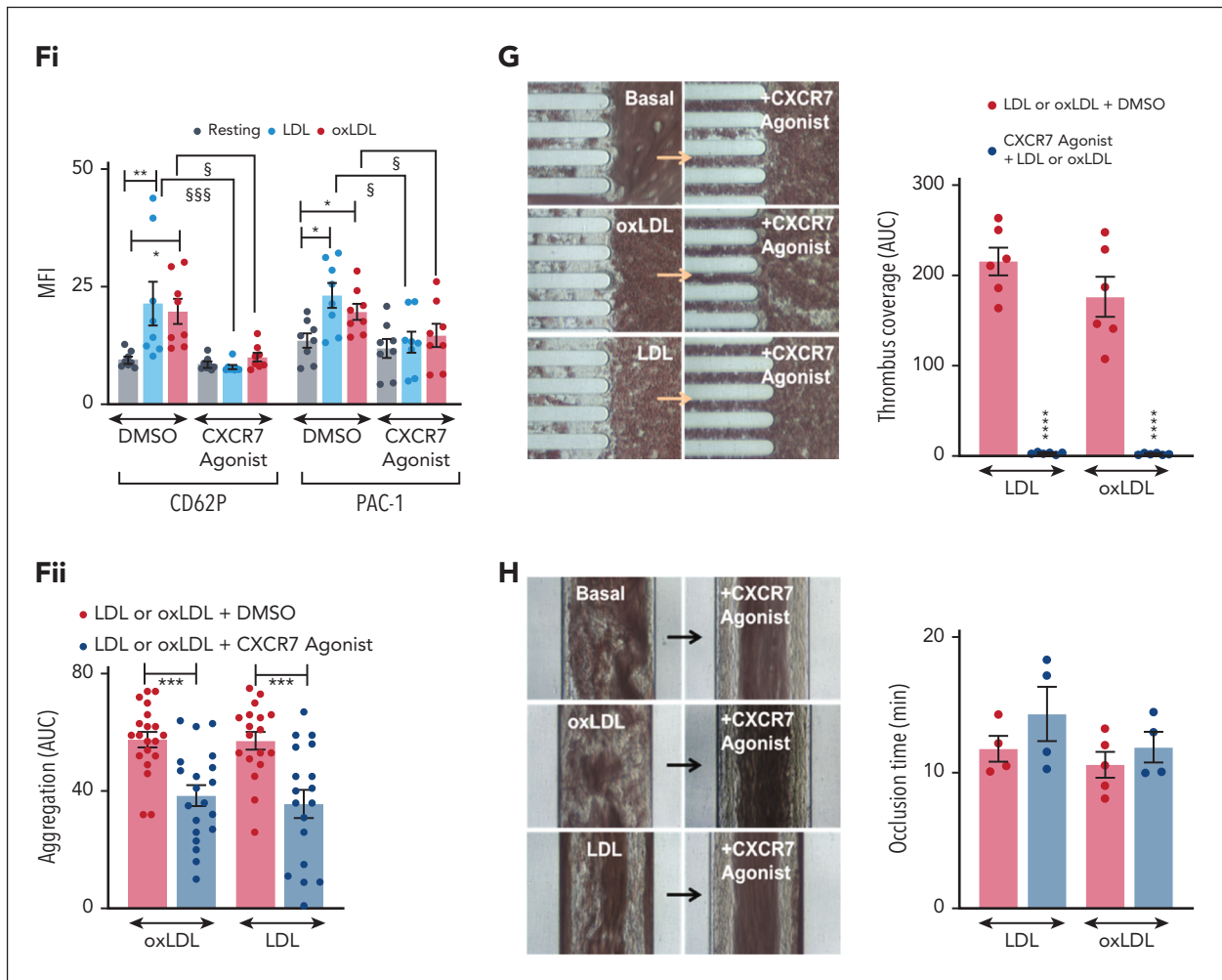


Figure 1 (continued) or CXCR7 agonist VUF11207 (100 $\mu\text{g}/\text{mL}$), or iloprost (20nM), or the cell-permeable SOD mimetic MnTMPyP (100 μM) for 30 minutes at room temperature, followed by stimulation with platelet-activating CRP (5 $\mu\text{g}/\text{mL}$) or LDL (20 $\mu\text{g}/\text{mL}$) for 60 minutes at room temperature. Intraplatelet oxidation of LDL to oxLDL was subsequently detected using a FITC-conjugated anti-human oxLDL antibody optimized for flow cytometry after platelet fixation (1% paraformaldehyde) and permeabilization (0.3% Triton X-100), accumulating fluorescence signals for platelets (FSC-SSC) by flow cytometry. (A) Intraplatelet oxidation of LDL to oxLDL in CRP- and LDL-activated platelets treated with CXCR7 agonist (100 $\mu\text{g}/\text{mL}$), iloprost (20nM), or MnTMPyP (100 μM), or vehicle control. Data are presented as mean $\pm$ standard error of the mean (SEM) from $n = 5$ experiments with experiments performed in duplicates. **** $P < .0001$; *** $P = .0001$ by analysis of variance (ANOVA) and Šidák multiple-comparisons test. For detection of intraplatelet lipid peroxidation, PRP platelets (1×10^6 per sample) were preloaded with the lipid peroxidation sensor C11-BODIPY^{581/591} (5 μM) for 15 minutes, followed by treatment with the CXCR7 agonist VUF11207 (100 $\mu\text{g}/\text{mL}$) or vehicle control (1% DMSO) for 30 minutes at room temperature. Thereafter, platelets were either kept under resting condition or treated with LDL (20 $\mu\text{g}/\text{mL}$), oxLDL (20 $\mu\text{g}/\text{mL}$), or CRP (5 $\mu\text{g}/\text{mL}$) for 60 minutes to induce lipid peroxidation and analyzed by flow cytometry. (Bi) Intraplatelet lipid peroxidation assessed by the C11-BODIPY^{581/591} fluorescence response with flow cytometry after CRP-, LDL-, or oxLDL-mediated platelet activation in the presence or absence of the CXCR7 agonist VUF11207 (100 $\mu\text{g}/\text{mL}$) or vehicle control (DMSO). Data are presented as mean $\pm$ SEM. **** $P < .0001$; *** $P = .0004$ by 1-way ANOVA with Šidák post hoc test. In a separate set of experiment, isolated platelets were incubated with the CXCR7 agonist VUF11207 (50 $\mu\text{g}/\text{mL}$) or DMSO (0.1%) for 20 minutes and then with C11-BODIPY^{581/591} (5 μM) for 10 minutes. OxLDL (20 $\mu\text{g}/\text{mL}$) or LDL (20 $\mu\text{g}/\text{mL}$) was added and incubated for 20 minutes. Thereafter, platelets were allowed to adhere to a fibrinogen-coated (100 $\mu\text{g}/\text{mL}$) glass-bottom cell culture dish for 15 minutes. Fluorescence images were acquired immediately using an inverted microscope (Nikon Ti-Eclipse) equipped with a 100 $\times$ objective (Plan Apo λ 100 $\times$ Oil Ph3 DM). Platelet MFI was quantified using the particle analysis function of the FIJI image-processing package of the Image J software (supplemental Figure 2A). (Bii) Quantitative analysis from C11-BODIPY^{581/591} fluorescence images demonstrating lipid peroxidation in LDL- or oxLDL-activated platelets in the presence or absence of the CXCR7 agonist VUF11207 or vehicle control (DMSO). Data are presented as mean $\pm$ SEM of individual platelets from $n = 5$ experiments. **** $P < .001$ by ANOVA with Šidák post hoc test. PRP platelets (1×10^6 per sample) from healthy donors were pretreated with vehicle control (1% DMSO), or CXCR7 agonist (100 $\mu\text{g}/\text{mL}$), or iloprost (20nM), or MnTMPyP (100 μM) for 30 minutes at room temperature and then either kept under resting condition or treated with LDL (20 $\mu\text{g}/\text{mL}$) or oxLDL (20 $\mu\text{g}/\text{mL}$) for 1 hour at room temperature. $\Delta\psi_{\text{m}}$ change was evaluated by decrease in TMRE fluorescence using flow cytometry. (C) Mitochondrial membrane depolarization in LDL- or oxLDL-activated platelets in the presence or absence of CXCR7 agonist, iloprost, MnTMPyP, or vehicle control (DMSO). Data are presented as mean $\pm$ SEM from 7 independent experiments performed in duplicates. **** $P < .0001$; *** $P = .0004$; ** $P = .005$; * $P = .01$ by ANOVA with Šidák post hoc test. PRP platelets (1×10^6 per sample) from healthy donors were preloaded with mitochondrial superoxide detecting MitoSOXRed (3 μM) for 15 minutes at room temperature. Thereafter, platelets were treated with vehicle control (1% DMSO), the CXCR7 agonist VUF11207 (100 $\mu\text{g}/\text{mL}$), iloprost (20nM), or MnTMPyP (100 μM) for 30 minutes at room temperature. Platelets were then either kept under resting condition or treated with LDL (20 $\mu\text{g}/\text{mL}$) or oxLDL (20 $\mu\text{g}/\text{mL}$) for 60 minutes at room temperature before analysis by flow cytometry. (D) Mitochondrial superoxide generation in LDL- or oxLDL-activated platelets in the presence or absence of the CXCR7 agonist VUF11207, iloprost, MnTMPyP, or vehicle control (DMSO). Data are presented as mean $\pm$ SEM from 10 experiments performed in duplicates. **** $P < .0001$ by ANOVA with Šidák post hoc test. (E) Exemplary SICM topography images (top), spreading area (bottom left), and cell height (bottom right) of human platelets spread on fibrinogen- or collagen-coated surfaces after preincubation with CXCR7 agonist or vehicle control and activation with oxLDL or LDL. Data are presented as arithmetic mean $\pm$ SEM of $n = 24$ to 27 platelets from 3 donors for each condition. * $P < .05$; ** $P < .01$; *** $P < .001$ by t test. (Fi) Platelet CD62P surface expression induced by LDL (20 $\mu\text{g}/\text{mL}$) or oxLDL (20 $\mu\text{g}/\text{mL}$) and PAC-1 binding denoting $\alpha_{\text{IIb}}\beta_{3}$ -integrin activation in blood from healthy donors in the presence or absence of the CXCR7 agonist VUF11207 (100 $\mu\text{g}/\text{mL}$) or vehicle control (DMSO). Data are presented as mean $\pm$ SEM. * $P < .05$; ** $P < .01$; § $P < .05$; §§§ $P < .01$ using paired t test. (Fii) Aggregation response

Lipids are consumed through mitochondrial β -oxidation to meet the increased energy demands for functional response.^{3,56} Upon thrombin stimulation, generation of the signaling intermediate DG was significantly reduced by the inhibitory action of CXCR7 agonist (Figure 2Bii). Nevertheless, we observed a significantly increased accumulation of LC-CAR in CXCR7 agonist-treated thrombin-activated platelets (Figure 2Biv), owing to persistent lipolysis that generated free fatty acids (FAs) from lipid reserves (TGs and CEs), which were subsequently processed into LC-CAR by CPT1. This effect of CXCR7 agonist was unique and significantly distinct from that of iloprost, which did not influence LC-CAR levels in resting or thrombin-activated platelets (supplemental Figure 4), despite a similar stabilizing effect on $\Delta\psi_m$ and in preventing mitochondrial superoxide generation (Figure 1C-D).

LC-CAR binds to and inhibits FXa-mediated thrombin generation and clotting, as demonstrated in a biochemically reconstituted cell-free coagulation system using recombinant coagulation factors and LC-CAR supplementation.¹⁶ Levels of LC-CAR are reduced in patients with STEMI,⁵⁷ VTE,¹⁶ and VITT.¹⁸ To ascertain the relevance of CXCR7 agonist-induced intraplatelet LC-CAR reserve, we estimated their accumulation under the influence of CXCR7 agonist and subsequent release from platelets, particularly focusing on anticoagulant LC-CAR.^{16,58,59} We observed a substantial increment in intraplatelet LC-CAR levels (16:0, 18:0, 18:1, and 18:2) (Figure 3A-B) and a corresponding release of LC-CAR 18:1 and LC-CAR 18:2 into the platelet supernatant (Figure 3A-B) upon CXCR7 agonist treatment in both resting and thrombin-activated platelets. Therefore, CXCR7 agonist-induced platelet LC-CAR may replenish circulating anticoagulant LC-CAR in patients, thereby counteracting their procoagulant disposition.¹⁶

Moreover, elevated levels of LC-CAR in the supernatant of CXCR7 agonist-treated platelets (Figure 3A-B) could also drive CD36 internalization. FAT/CD36 can transport LC-CAR^{48,50,51} into mitochondria for subsequent oxidation, which is counteracted by the CD36 inhibitor sulfo-*N*-succinimidyl oleate.^{48,60} Accordingly, FAT/CD36 can be trafficked directly from the plasma membrane or intracellular pools into mitochondria.^{51,60,61}

Dual impact of CXCR7 agonist on platelet-assisted coagulation

We verified the inhibitory effect of CXCR7 agonist on CRP-A-induced procoagulant platelet transformation (CD62P⁺ and PS-externalizing lactadherin⁺ or annexin V⁺ platelets⁶²) (Figure 3Ci), like iloprost,^{63,64} and consequently observed significantly reduced FX/FXa binding to CRP-A- and convulxin plus TRAP6-activated platelets (Figure 3Cii). To gain a physiological perspective on the potential implication of platelet-released LC-CAR, we tested procoagulant platelet-assisted, FXa-dependent thrombin generation in PRP under the influence of CXCR7 agonist using a calibrated automated

thrombography assay rather than a cell-free biochemically reconstituted clotting system,¹⁶ where all coagulation factors, including FXa and prothrombin, are present in plasma, whereas LC-CARs are derived from CXCR7 agonist-treated platelets (Figure 3D). Therefore, we used the CPT1 inhibitor etomoxir to prevent the generation of anticoagulant LC-CARs. CXCR7 agonist did not affect basal/resting platelet-assisted thrombin generation in PRP or PPP, as previously observed¹⁴ (Figure 3D). CRP-induced procoagulant platelet-assisted thrombin generation in PRP was significantly reduced by CXCR7 agonist (Figure 3D), attributable to reduced PS exposure (Figure 3Ci) and FX/FXa binding (Figure 3Cii). However, this inhibitory effect of CXCR7 agonist on platelet-assisted thrombin generation in PRP was partly ($P = .0376$) reversed by etomoxir (Figure 3D), preventing the generation and release of LC-CARs from platelets in PRP without exerting any effect in the cell-free (CPT1-free) PPP system. The counteracting effect of etomoxir revealed the influence of platelet-derived anticoagulant LC-CAR in the process. LC-CAR(16:0) binds to FXa and interferes with thrombin generation, as demonstrated in a biochemically reconstituted system using LC-CAR(16:0) and chemically modified FXa in surface plasmon resonance binding studies.¹⁶ Lipidomics analysis cannot assess LC-CAR-FXa complexes already formed in PRP because proteins are precipitated during lipid extraction for liquid chromatography-tandem mass spectrometry analysis. To ascertain platelet-derived LC-CAR binding to plasma FXa and the resultant inhibitory effect in a physiological system like PRP, we relied on functional response (reduced thrombin generation) and its reversal when LC-CAR generation was prevented by the CPT1 inhibitor etomoxir. Thus, CXCR7 agonist may exert a 2-pronged effect on procoagulant platelet-assisted thrombin generation, first, by counteracting PS externalization and thereby preventing coagulation factor assembly and, second, unlike iloprost, by inducing the generation and release of anticoagulant LC-CARs, which can affect FXa-dependent generation of thrombin in the immediate microenvironment (Figure 3D; schematic presentation).

Gaining a translational, clinical perspective, we observed that intraplatelet levels of anticoagulant CARs, TGs, DGs, CEs, sphingomyelins, and ceramides (Figure 4A; supplemental Table 1) in patients with CAD ($n = 107$) were not affected by antiplatelet therapy, concomitant HL, diabetes mellitus, or disease severity (ACS vs CCS). However, ex vivo treatment of CAD platelets with CXCR7 agonist significantly ($P = .001$; $q = 0.0338$) increased anticoagulant LC-CAR levels (16:0, 18:0, and 18:1) in thrombin-activated platelets to greater than twofold and in resting platelets to greater than two- to fourfold ($P = .001$; $q = 0.0593$) (Figure 4B; supplemental Table 2),²⁹ which could regulate platelet-assisted coagulation. Nevertheless, the accumulation of intraplatelet CARs could also stem from alterations in carnitine shuttle (Figure 4C) and mitochondrial metabolism, which was verified.

Figure 1 (continued) to LDL (20 $\mu\text{g/mL}$) or oxLDL (20 $\mu\text{g/mL}$) ($n = 5-8$) in the presence or absence of CXCR7 agonist (100 $\mu\text{g/mL}$) or vehicle control (DMSO). Data are presented as mean $\pm$ SEM. *** $P < .001$ by ANOVA with Šidák multiple-comparisons test. Phase contrast still images and T-TAS data ($n = 4-6$) showing thrombus coverage (AUC) over collagen-coated (PL chip) surface (G) and over collagen plus TF-coated (AR chip) surface (H) at arterial shear rates. Thrombus coverage in PL chip (G) and time to AR chip occlusion (H) in the presence or absence of the CXCR7 agonist VUF11207 (100 $\mu\text{g/mL}$) or vehicle control (DMSO). Data are presented as mean $\pm$ SEM. * $P < .05$ and **** $P < .001$ by ANOVA with Šidák multiple-comparisons test. AUC, area under the curve; FITC, fluorescein isothiocyanate; MFI, mean fluorescence intensity; SOD, superoxide dismutase; SICM, scanning ion conductance microscopy; TMRE, tetramethylrhodamine ethyl ester.

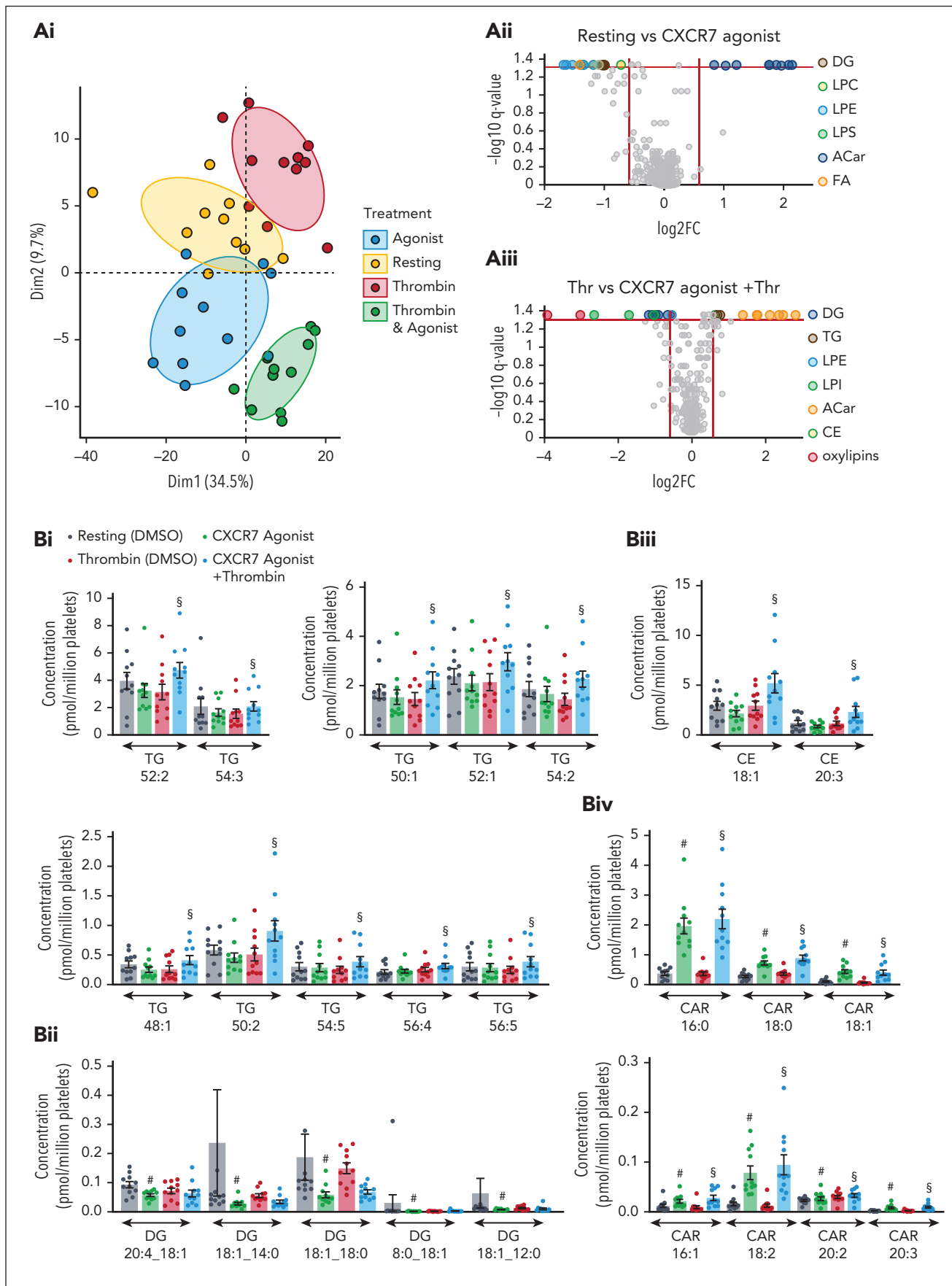


Figure 2.

CXCR7 preserves platelet energy status and sustains mitochondrial metabolism

We assessed the influence of CXCR7 agonist on real-time mitochondrial respiration oxygen consumption rate (OCR) and extracellular acidification rate (ECAR)³⁵ in resting and thrombin-activated platelets (Figure 5). Thrombin-induced activation significantly increased OCR irrespective of the presence of CXCR7 agonist (Figure 5A). Moreover, both treatments with thrombin and thrombin plus CXCR7 agonist increased ATP-linked respiration (Figure 5B). Although maximal respiration was considerably decreased by CXCR7 agonist under resting conditions ($P = .004$), consistent with its inhibitory influence on platelet activity,¹⁴ it increased maximal respiration in thrombin-activated platelets ($P = .02$; Figure 5B), supporting oxidative metabolism. Spare respiratory capacity, which indicates how cells respond to increased energy demand, was decreased by CXCR7 agonist in resting platelets, which have lower energy demand, but was significantly improved in thrombin-activated platelets (Figure 5B). Prolonged (90 minutes) stimulation (Figure 5A) with thrombin during OCR ECAR measurements could have caused gradual metabolic breakdown and $\Delta\psi_m$, the latter of which is counteracted by both physiological (CXCL11, MIF, and CXCL12)^{33,65} and pharmacological (VUF11207) CXCR7 ligands, thereby preserving mitochondrial integrity (Figure 1C-D) and functionality. Consequently, maximal respiration and spare respiratory capacity were increased in platelets treated with thrombin plus CXCR7 agonist compared to thrombin-treated platelets.

The differential impact of CXCR7 agonist on the mitochondrial respiratory chain could also result from preferential use of energy substrates in resting and thrombin-activated platelets.^{56,66} Therefore, we explored the use of different metabolic fuels for mitochondrial respiration using the Mito Fuel test (supplemental Figure 5). For all 3 possible mitochondrial fuels (glucose, glutamine, and LCFAs), substrate dependency was significantly higher, whereas flexibility was markedly lower in thrombin-activated platelets than in resting platelets irrespective of the presence of CXCR7 agonist. Moreover, FA and glucose capacity were lower in activated platelets and remained unchanged for glutamine (supplemental Figure 5A-C). These results indicate that CXCR7 agonist-induced preferential accumulation of LC-CAR was not because of a disruption of mitochondrial metabolism.

To gain further insights into mitochondrial metabolism under the influence of the CXCR7 agonist VUF11207, we assessed the platelet metabolic profile by targeted and untargeted metabolomics analysis (Figure 6; supplemental Figures 6 and

7; supplemental Table 3). Thrombin activation significantly reduced ADP and ATP levels (Figure 6A), attributable to granule release and energy consumption associated with degranulation. The significant increase in energy charge (Figure 6A) was consistent with previous reports demonstrating the presence of functional, active mitochondrial oxidative phosphorylation.^{67,68} This interpretation is further supported by the observed increase in mitochondrial ATP-linked OCR and acute OCR in thrombin-activated platelets compared to resting controls (Figure 5B). In resting platelets, CXCR7 agonist improved ATP/ADP ratio and energy charge⁶⁷ (Figure 6A), confirming sustained basal metabolism despite its inhibitory effect on platelet response. ATP production from glucose metabolism involves glycolysis, the tricarboxylic acid cycle, and oxidative phosphorylation. Intracellular levels of central metabolites from these pathways, including glucose, pyruvate, lactate, and citrate, remained unaltered after CXCR7 agonist treatment in resting platelets (Figure 6B). The use of glutamine as a potential substrate for ATP production was also unaltered by CXCR7 agonist (Figure 6B). Thrombin-induced activation led to an expected increase in lactate levels compared to resting platelets (Figure 6B), consistent with their glycolytic phenotype.⁶⁶ CXCR7 agonist sustained glucose levels and counteracted thrombin-induced lactate generation, which might point to reduced glycolytic activity, combined with significantly increased citrate levels and increased input of glutamine into the tricarboxylic acid cycle (Figure 6B). Mitochondrial β -oxidation of FAs contribute to ATP production.⁵⁶ A pronounced increase in intraplatelet LC-CAR levels (Figures 2-4) after CXCR7 agonist treatment was confirmed by metabolomics analyses of resting and thrombin-activated platelets (Figure 6C; supplemental Figure 7) and a concomitant decrease in carnitine levels (Figure 6C), reflecting its consumption during LC-CAR generation. Interestingly, medium- and short-chain CARs were not increased by CXCR7 agonist (Figure 6C; supplemental Figure 6A-B), suggesting preferential channeling of LC-CAR through the mitochondrial carnitine shuttle for consumption by β -oxidation.⁵⁶ Eventually, CXCR7 agonist-induced accumulation of LC-CAR could have inhibited carnitine palmitoyl transferase 2 (CPT2)⁶⁹ thereby preventing the recycling of carnitine and causing a decline in intraplatelet carnitine levels (Figure 6C). Overall, metabolically active mitochondria in platelets under the influence of CXCR7 agonist preserved $\Delta\psi_m$ and prevented mitochondrial superoxide generation (Figure 1C-D), which otherwise results from electron leakage from the electron transport chain.

Preference for lipolysis over lipogenesis

Accumulation of LC-CAR suggested the generation of free-LCFAs after lipolysis of lipid reserves (Figure 3Bi-iii), which

Figure 2. CXCR7 ligation favors generation of anticoagulant CAR in platelets. Washed platelets (300×10^6 per sample) from healthy donors were pretreated with the CXCR7 agonist VUF11207 (100 $\mu\text{g/mL}$) or vehicle control (1% dimethyl sulfoxide [DMSO]) for 30 minutes at room temperature. Then, platelets were either kept under resting condition or activated with thrombin (0.1 U/mL) for 15 minutes at room temperature. Thereafter, platelets were processed for lipidomics analysis. (Ai) Unsupervised PCA for z-scaled data (for each donor) showing distinct clustering with respect to different experimental conditions. (Aii-iii) Volcano plots comparing the effect of CXCR7 agonist on resting and thrombin-activated (0.1 U/mL) platelets. The labeled points present lipids, levels of which are significantly changed as assessed by plots of q value vs FC. (B) Scatterplots with bars showing relative intraplatelet levels of TGs (i), DGs (ii), CEs (iii), and CARs (iv) in the presence or absence of CXCR7 agonist (100 $\mu\text{g/mL}$) or vehicle control (1% DMSO). Concentrations (picomoles per 10^6 platelets) were calculated using 1-point calibration with class-specific internal standards. For CARs, with no class-specific internal standard, LPC internal standard (IS) was used as a surrogate IS. Calculated concentrations were corrected using a response factor determined by slope surrogate/slope lipid after acquisition. Statistical comparisons were performed using the nonparametric paired Wilcoxon signed-rank test with FDR correction. The # denotes significant difference compared with resting platelets, and § denotes a significant difference compared with thrombin-activated platelets. FC, fold change; FDR, false discovery rate; PCA, principal component analysis.

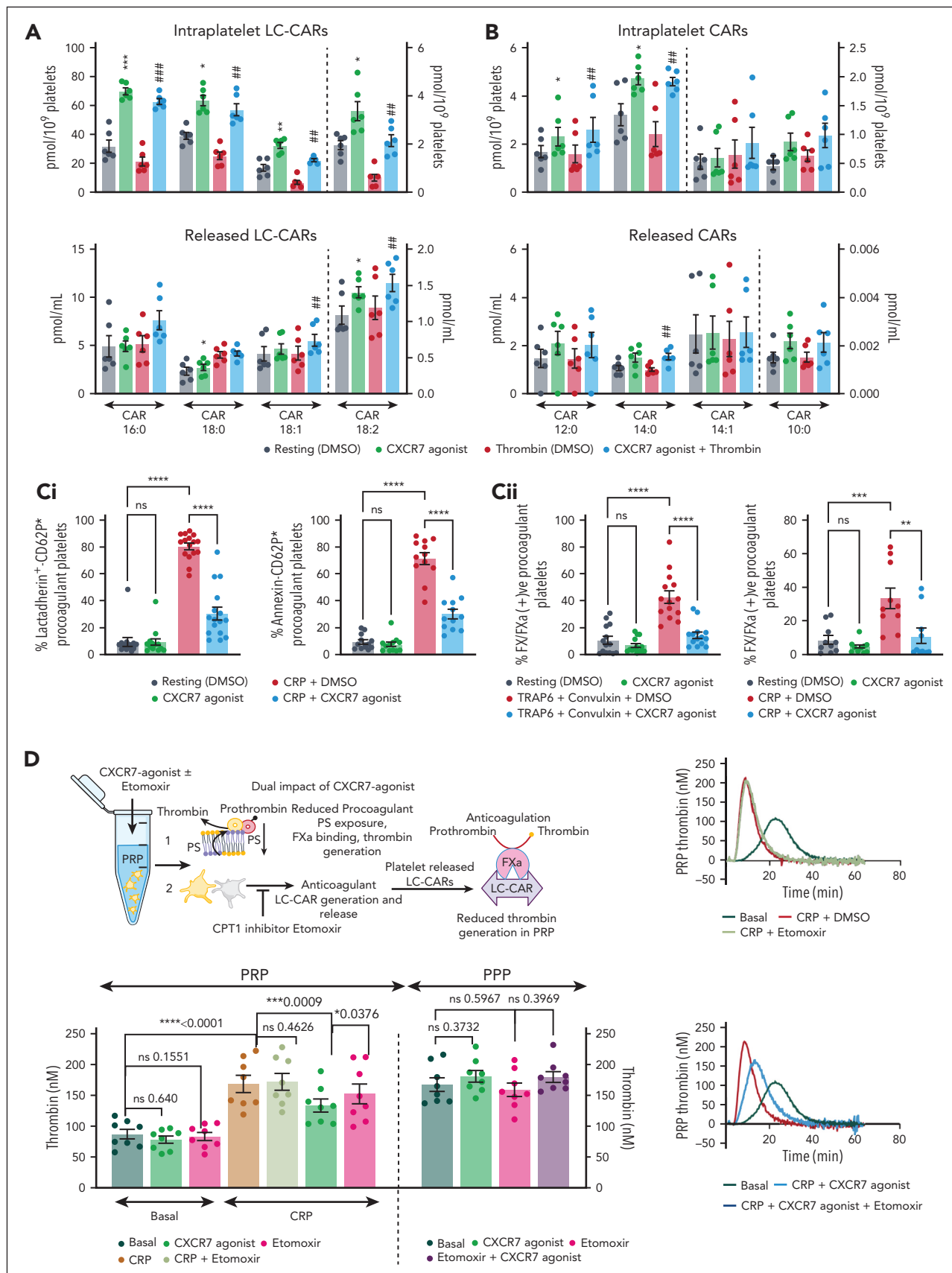


Figure 3. CXCR7 agonist-induced CARs may influence platelet-assisted thrombin generation. (A-B) Washed platelets (300×10^6 per sample) from healthy donors were pretreated with the CXCR7 agonist VUF11207 (100 $\mu\text{g}/\text{mL}$) or vehicle control (1% DMSO) for 30 minutes at room temperature. Then, platelets were either kept under resting condition or activated with thrombin (0.1 U/mL) for 15 minutes at room temperature. Thereafter, the samples were centrifuged at 600g for 5 minutes at 4°C. The

were converted to LC-CAR by CPT1 under the influence of CXCR7 agonist (Figure 2Biii; Figure 3A-B; Figure 4B). We deciphered the molecular basis behind this preferential lipolysis and subsequent LC-CAR generation. AMP-dependent kinase (AMPK) is directly regulated by adenylate charge,⁷⁰ which is improved by CXCR7 agonist (Figure 6B), thereby sustaining metabolism, besides, the AMPK-ACC pathway is a crucial sensor of energy status that regulates mitochondrial carnitine shuttle.⁷¹ CXCR7 ligation elevates cAMP levels in platelets by coordinating with the G_{α_s} -coupled IP receptor through 12-HETrE.¹⁴ cAMP is eventually hydrolyzed to AMP by phosphodiesterases. Therefore, we explored the influence of CXCR7 agonist on the AMPK-ACC axis. Without affecting AMPK^{Ser485/491} phosphorylation (Figure 6Di), CXCR7 agonist induced AMPK^{Thr172} phosphorylation (Figure 6Dii), causing its activation, and subsequently, AMPK-mediated inhibitory phosphorylation of ACC1^{Ser79} (Figure 6Diii), thereby preventing malonyl-CoA synthesis, which is essential for de novo lipogenesis. This demonstrates the preference of pharmacological CXCR7 agonists CCX771⁴⁵ and, currently, VUF11207 for lipolysis over lipogenesis.

CXCR7 regulates thromboinflammatory functions in venous thrombosis

Procoagulant platelets play a critical role in venous thrombosis, as demonstrated by their increased levels in the circulation and in thrombi of patients with deep vein thrombosis and pulmonary embolism.⁷² Since CXCR7 agonist regulated procoagulant platelet-assisted coagulation and prompted anticoagulant LC-CARs generation (Figures 2-4), we validated its influence on procoagulant and thromboinflammatory platelet functions in venous thrombosis. Administration of CXCR7 agonist (100 μ g per mouse IV, as in the arterial thrombosis model¹⁴) reduced thrombus formation in the IVC (Figure 7A) ligated for 48 hours and significantly regulated $\alpha_{IIb}\beta_{III}$ -integrin activation (JON/A

binding) (Figure 7Bi), degranulation (CD62P) (Figure 7Bii), thromboinflammatory platelet-leukocyte aggregate formation (Figure 7Ci), and procoagulant platelet activity (Figure 7Cii) under basal circulating conditions and after ex vivo stimulation with CRP. Lipidomics analysis of murine plasma showed significantly reduced levels of thrombotic oxylipins, including platelet-derived 12-HETE (12-LOX), 12-KHT (COX-1 pathway), and leukocyte-derived 15-HETE (5-/15-LOX) (Figure 7Di), which could otherwise generate procoagulant HETE-PLs that are richly deposited in limb, carotid, and aortic thrombi.¹³ Conversely, increased levels of anticoagulant LC-CARs (supplemental Figure 8) were observed, circulating levels of which are reduced in patients with VTE.¹⁶ Therefore, CXCR7 agonist exerted a potential therapeutic effect against procoagulant platelets by reducing PS exposure on circulating platelets (Figure 7Cii), preventing the possibility of procoagulant HETE-PL generation (Figure 7Di), and inducing anticoagulant LC-CARs in circulation (supplemental Figure 8), corresponding to reduced thrombus burden (Figure 7A). These results also demonstrate the ability of CXCR7 agonist to reduce enzymatic lipid peroxidation (12-HpETE, 5-HpETE, and 15-HpETE through 12-/5-/15-LOX), similar to its effects on nonenzymatic ROS-mediated lipid peroxidation (Figure 1Bi-ii).

Plasma levels of platelet- and leukocyte-derived inflammatory mediators, including IL-1 β , IL-6, interferon gamma, TNF- α , and MCP-1, were decreased in CXCR7 agonist-treated mice (Figure 7Dii). Therefore, systemic administration of CXCR7 agonist may also exert an influence on circulating leukocytes to reduce thromboinflammation, corroborating our previously documented findings that CXCR7 agonist VUF11207 restricts leukocyte infiltration into the infarcted myocardium in vivo¹⁵ and reduces neutrophil CD11b activation and platelet-neutrophil aggregate formation otherwise induced by sera/IgG from patients with heparin-induced thrombocytopenia.¹⁴

Figure 3 (continued) supernatants were separated from the platelet pellets, and both were used for lipid extraction and subsequent lipidomics analysis. LC-CAR generation in platelets and release of the same LC-CAR in resting and thrombin-activated platelet supernatant are graphically presented in (A-B). Data are presented as mean $\pm$ standard error of the mean (SEM) derived from experiments done with platelets from $n = 6$ healthy donors. FDR-adjusted P values indicate * $P < .05$ and *** $P < .0001$ in platelets treated with the CXCR7 agonist VUF11207 (100 μ g/mL) as compared to resting platelets (plus vehicle control DMSO); ## $P < .001$ and ### $P < .0001$ in the presence of CXCR7 agonist as compared to thrombin (plus vehicle control DMSO) treatment alone. (C) PRP platelets (1×10^6 per sample) from healthy donors were pretreated with 1% DMSO or CXCR7 agonist (100 μ g/mL) for 30 minutes at room temperature and then kept under resting condition or treated with dual platelet agonists convulxin (0.1 μ g/mL) plus TRAP-6 (10 μ M) or with CRP-A (5 μ g/mL) for 1 hour at room temperature. PS exposure was ascertained as an increase in annexin V-fluorescein isothiocyanate (FITC) response and lactadherin-FITC binding, together with surface expression of CD62P evaluated by anti-CD62P APC antibody. (Ci) Bar diagram representing the percentage of procoagulant platelets induced by CRP-A stimulation in the presence or absence of CXCR7 agonist or vehicle control DMSO. Data are presented as mean $\pm$ SEM from PRP obtained from $n = 6$ to 8 healthy donors. *** $P < .0001$ by analysis of variance (ANOVA) with Šidák post hoc test. Binding of coagulation factor FX/FXa on procoagulant platelets was evaluated with mouse anti-human FX/FXa antibody, followed by counterstaining with Alexa Fluor 488-conjugated secondary antibody. (Cii) Percentage of FX/FXa binding procoagulant platelets as induced by CRP-A and dual agonist treatment of platelets in the presence or absence of the CXCR7 agonist VUF11207 or vehicle control (DMSO). Data are presented as mean $\pm$ SEM derived from experiments done with PRP platelets from $n = 5$ to 7 healthy donors. *** $P < .0001$ by ANOVA with Šidák post hoc test. (D) Schematic diagram showing the dual impact of CXCR7 agonist in platelet-assisted thrombin generation in PRP. First, CXCR7 agonist reduces procoagulant transformation of activated platelets by regulating PS exposure and FXa assembly. Second, it increases accumulation of LC-CARs in platelets, which upon release interfere with FXa-driven thrombin generation. PRP and PPP samples were pretreated with the CAR-generating enzyme CPT1 inhibitor etomoxir (40 μ M) or vehicle control for 15 minutes, followed by the CXCR7 agonist VUF11207 (100 μ g/mL) or vehicle control (1% DMSO) for an additional 30 minutes at room temperature. Etomoxir was used to prevent CXCR7 agonist-induced CAR generation in platelets and their subsequent release in PRP samples, whereas cell-free (and, therefore, CPT1-free) PPP samples were used in parallel as a control, in which etomoxir did not exert any effect in the absence of the intracellular target CPT1. After specific experimental treatments, samples were added to TF-containing wells in the presence or absence of CRP (10 μ g/mL). By adding the fluorogenic thrombin substrate Z-Gly-Gly-Arg aminomethyl coumarin (2.5 mM), coagulation reaction was started, and FXa-dependent thrombin generation was assessed by calibrated automated thrombinography. Bar graphs show thrombin generation in PRP and PPP in the presence or absence of CXCR7 agonist with or without the CPT1 inhibitor etomoxir. Data are presented as mean $\pm$ SEM derived from experiments done with PRP and PPP from $n = 8$ healthy donors. Statistical significance was assessed using a paired t test. Representative thrombinograms showing thrombin generation in PRP and PPP samples under different experimental conditions are shown. APC, allophycocyanin; FDR, false discovery rate.

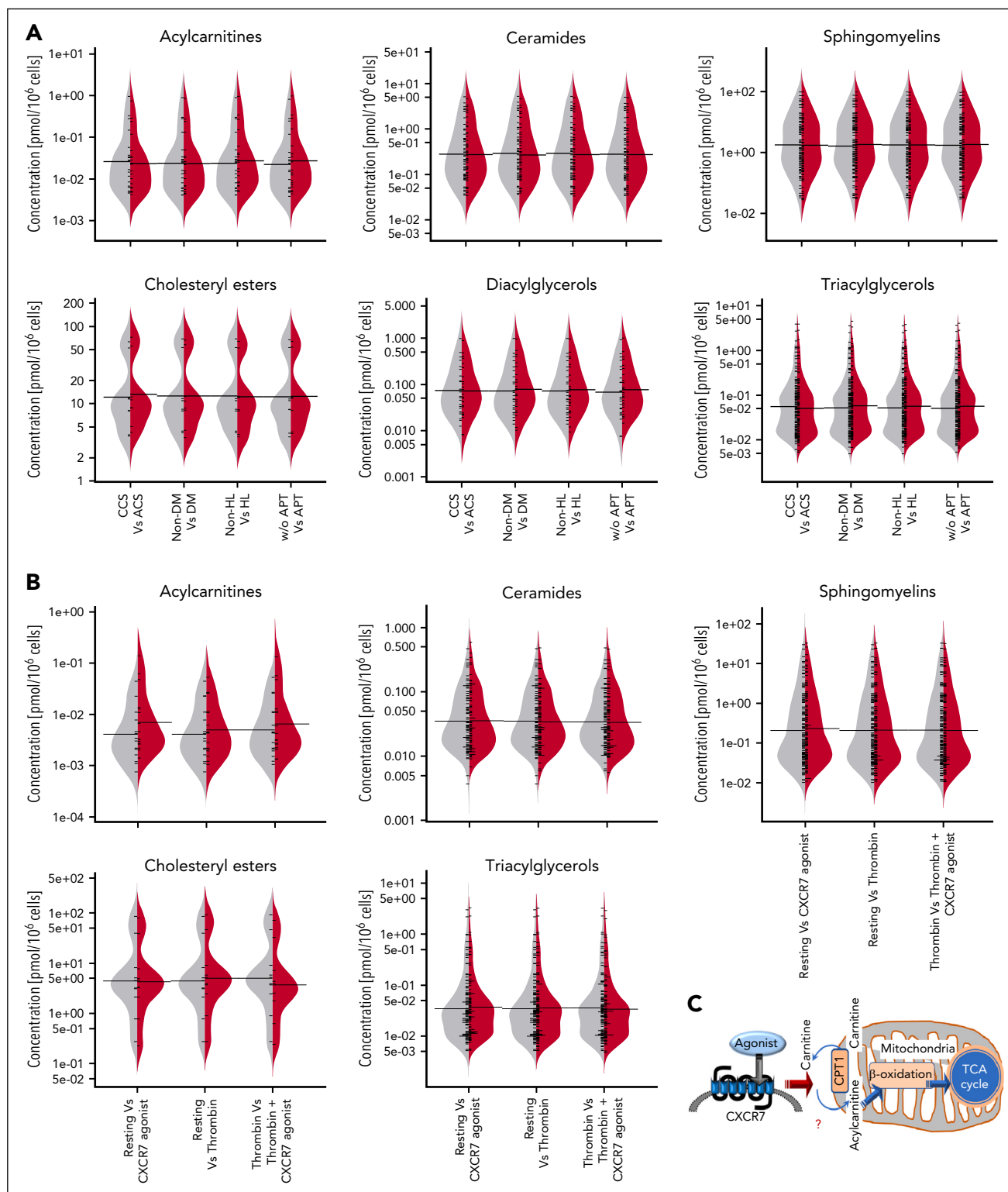


Figure 4. CXCR7 ligation favors CAR generation in platelets from patients with CAD. Asymmetric bean plots showing lipid species from different lipid classes, including CAR, CE, Cer, DG, SM, and TG, detected in untargeted platelet lipidomics. The small tick marks in the bean plots represent the average concentration of specific lipid species within each lipid class in a particular group (eg, CCS, ACS, diabetes mellitus [DM], hyperlipidemia [HL], antiplatelet therapy [APT]), and the large marks indicate the average concentration of a lipid class within a group (eg, CCS, ACS, DM, HL, APT). Concentrations are calculated using 1-point calibration with class-specific internal standard. (A) Data from patients with CAD ($n = 107$) are shown. Lipid levels were compared for CCS ($n = 81$) vs ACS ($n = 26$), non-DM ($n = 39$) vs DM ($n = 31$), non-HL ($n = 39$) vs. HL ($n = 52$), and patients without APT ($n = 46$) vs. with APT ($n = 61$). Washed platelet suspensions isolated from patients with CAD ($n = 15$) were pretreated with the CXCR7 agonist VUF11207 (100 $\mu\text{g}/\text{mL}$) or vehicle control (1% dimethyl sulfoxide [DMSO]) for 30 minutes at room temperature. Then, platelets were kept under resting condition or activated with thrombin (0.1 U/mL) for 15 minutes at room temperature. Afterward, the samples were snap-frozen in liquid nitrogen and stored at -80°C until untargeted lipidomics analysis. (B) Lipid concentrations of platelets from $n = 15$ CAD donors in resting (plus DMSO) condition and after ex vivo thrombin (plus DMSO)-induced activation (0.1 U/mL), CXCR7 agonist treatment (100 $\mu\text{g}/\text{mL}$), or thrombin-induced activation with CXCR7 agonist pretreatment are presented. (C) Schematic representation suggesting the potential influence of CXCR7 agonist on carnitine shuttle and mitochondrial metabolism in platelets. Cer, ceramide; SM, sphingomyelins.

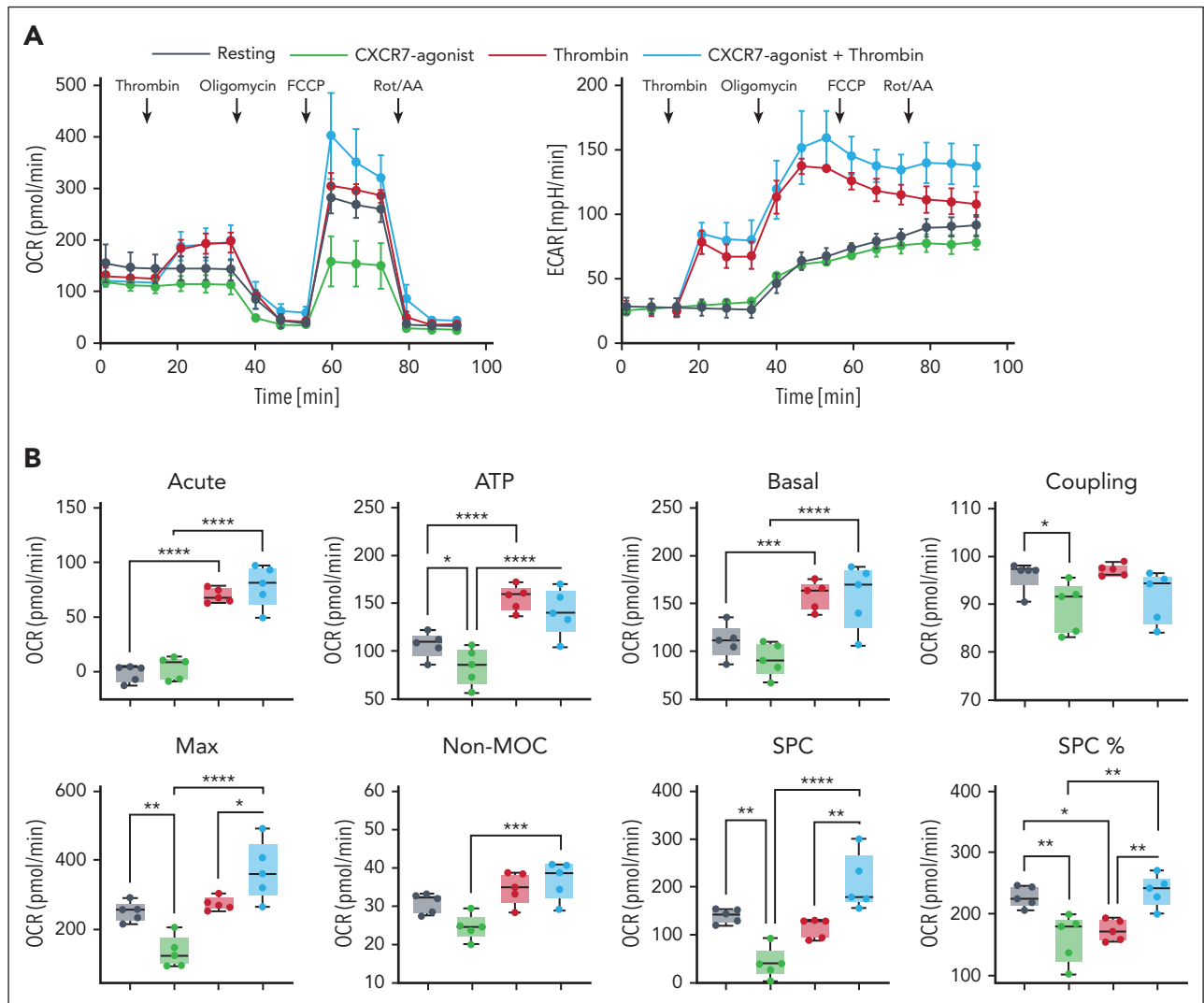


Figure 5. Mitochondrial function in platelets under the influence of CXCR7 agonist. Mitochondrial function was assessed by measuring OCR in platelets ($n = 5$ donors) in either resting condition or after thrombin-induced activation with or without the CXCR7 agonist VUF11207 (100 $\mu\text{g/mL}$) or DMSO pretreatment for 30 minutes. The Seahorse XF Cell Mito Stress Test kit (Agilent) was used according to the manufacturer's protocol, employing 15×10^6 platelets per well to assess mitochondrial respiration. Metabolic flux analysis was performed with an acute injection of the medium (resting platelets) or thrombin (0.1 U/mL) followed by sequential injection of oligomycin (1.1 μM), carbonyl cyanide 4-(trifluoromethoxy)phenylhydrazone (FCCP; 1 μM), and rotenone/antimycin A (Rot/AA) (0.5 μM). (A) Profiles of OCR and ECAR in response to modulators of respiration (oligomycin, FCCP, and Rot/AA). Mean values of 5 different donors are displayed (platelets from each donor were analyzed at least in triplicates). (B) Acute response of platelets to thrombin activation and key mitochondrial functional parameters, including ATP production, basal respiration, coupling efficiency, maximal respiration, nonmitochondrial respiration, and SRC. For each parameter, box-and-whiskers plots (minimum to maximum, with all data points shown) and mean values from 5 different donors are presented; each donor was analyzed at least in triplicate. Repeated-measures ANOVA was used to calculate differences in key parameters, followed by Šidák correction for multiple comparisons of selected pairs: resting vs CXCR7 agonist, resting vs thrombin, CXCR7 agonist vs thrombin plus CXCR7 agonist, thrombin vs thrombin plus CXCR7 agonist ($*P \leq .05$; $**P \leq .01$; $***P \leq .001$; $****P \leq .0001$). ANOVA, analysis of variance; DMSO, dimethyl sulfoxide; ECAR, extracellular acidification rate; OCR, oxygen consumption rate; SPC, spare respiratory capacity.

The inflamed, hypercoagulable vascular endothelium contributes significantly to circulating mediators in venous thrombosis, a process further aggravated by hyperlipidemia because lipoproteins activate endothelial cells.⁷³ Therefore, we explored the effect of VUF11207 on CXCR7-expressing endothelial cell-derived thrombotic/coagulant mediators from HUVEC under hyperlipidemic (LDL or oxLDL) conditions. CXCR7 agonist reduced the release of thromboinflammatory (sP-selectin, IL-6, sCD40L) and procoagulant TF mediators (Figure 7Ei-iii) from HUVECs, otherwise induced by lipoproteins. Thus, a regulation on platelet/leukocyte-derived procoagulant oxylipins (5-, 12-, 15-HETE) (Figure 7Di) and endothelial mediators (Figure 7Ei-iii) validated the therapeutic potential of targeting CXCR7 to

attenuate thromboinflammation in venous thrombosis, as previously demonstrated in myocardial infarction (MI) and arterial thrombosis.¹⁴

Discussion

The novel findings and mechanistic basis underlying the functional effects described in this investigation are as follows: (1) pharmacological targeting of CXCR7 induced activation of the AMPK^{Thr-172}-ACC^{Ser-79} pathway, favoring lipolysis of lipid reserves over lipogenesis; (2) this resulted in increased accumulation of intraplatelet anticoagulant¹⁶ LC-CAR without affecting mitochondrial integrity and metabolism; (3) the

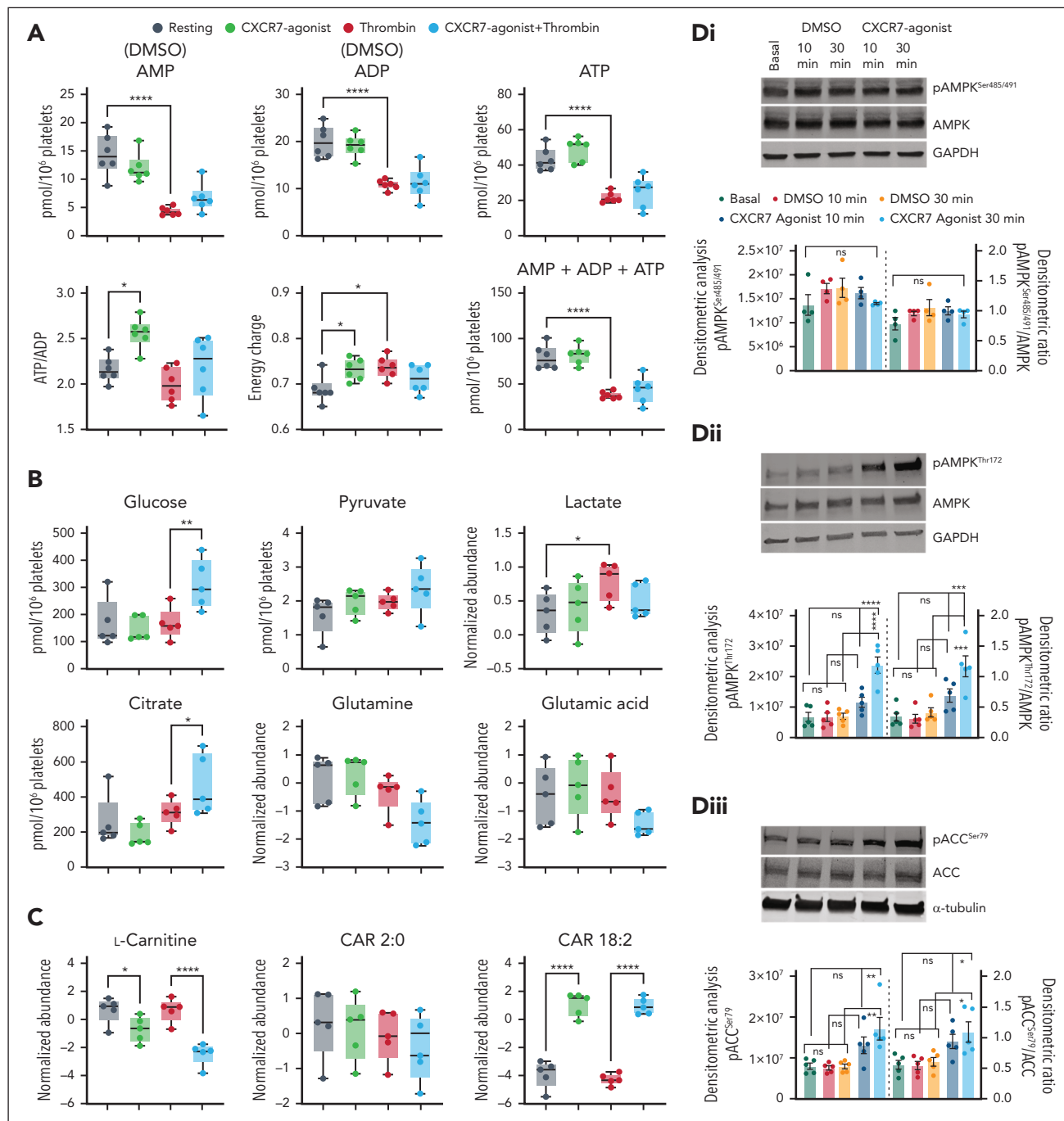


Figure 6. Metabolic profile of platelets and the metabolic sensor AMPK-ACC axis under the influence of CXCR7 agonist. Washed platelets (200×10^6 per sample) from $n = 6$ healthy donors were pretreated with the CXCR7 agonist VUF11207 (100 $\mu\text{g}/\text{mL}$) or vehicle control (1% dimethyl sulfoxide [DMSO]) for 30 minutes at room temperature. Then platelets were kept under resting condition or activated with thrombin (0.1 U/mL) for 15 minutes at room temperature. Thereafter, platelet pellets were collected for subsequent metabolomics analysis. Box-and-whisker plot (minimum to maximum, with all data points shown) showing intraplatelet levels of (A) AMP, ADP, ATP, ATP/ADP, and energy charge; (B) glucose, pyruvate, lactate, citrate, glutamine, and glutamic acid; (C) carnitine, acetylcarnitine, and LC-CAR 18:2 in samples of resting platelets and after thrombin-induced activation in the presence or absence of CXCR7 agonist with respect to treatment with vehicle control (DMSO). Calculation of concentrations (picomoles per 10^6 platelets) was done for AMP, ADP, ATP, glucose, pyruvate, and citrate, whereas for metabolites analyzed by profiling (carnitine, CARs, lactate, glutamine, and glutamic acid), normalized abundance is displayed. The energy charge was calculated as follows: $([\text{ATP}] + 0.5 \times [\text{ADP}]) / ([\text{ATP}] + [\text{ADP}] + [\text{AMP}])$.⁶⁷ Data are from $n = 5$ ($n = 6$ for ATP, ADP, and AMP) different donors (each donor sample was analyzed in duplicate). * $P \leq .05$; ** $P \leq .01$; *** $P \leq .001$; **** $P \leq .0001$ by repeated-measures analysis of variance (ANOVA) followed by Šidák correction for multiple comparisons of selected pairs: resting (DMSO) vs CXCR7 agonist, resting (DMSO) vs thrombin (DMSO), and thrombin (DMSO) vs thrombin plus CXCR7 agonist. (D) Isolated washed human platelets were treated with the CXCR7 agonist VUF11207 (100 $\mu\text{g}/\text{mL}$) or vehicle control (DMSO) for 10 and 30 minutes, respectively, at room temperature. A separate experimental set was kept as basal platelet sample. Samples were processed for immunoblot analyses to detect AMPK, pAMPK (Ser-485/491, Thr-172), ACC, and pACC (Ser-79), with GAPDH and α -tubulin as loading controls. Representative immunoblots and respective densitometric analyses of immunoblot images showing time-dependent phosphorylation status of (Di-ii) AMPK phosphorylation sites Ser-485/491 (Di), Thr-172 (Dii) and (Diii) ACC-Ser-79 in response to the CXCR7 agonist VUF11207 over time (10 and 30 minutes) with respect to basal levels and vehicle control (10 and 30 minutes). Data are presented as mean $\pm$ SEM and derived from $n = 4$ to 5 independent immunoblot analyses performed with platelets from different healthy donors. * $P \leq .05$; ** $P = .003$; *** $P = .0004$; **** $P < .0001$ by ANOVA with Šidák correction for multiple comparisons, as indicated in the graphs. GAPDH, glyceraldehyde-3-phosphate dehydrogenase; SEM, standard error of the mean.

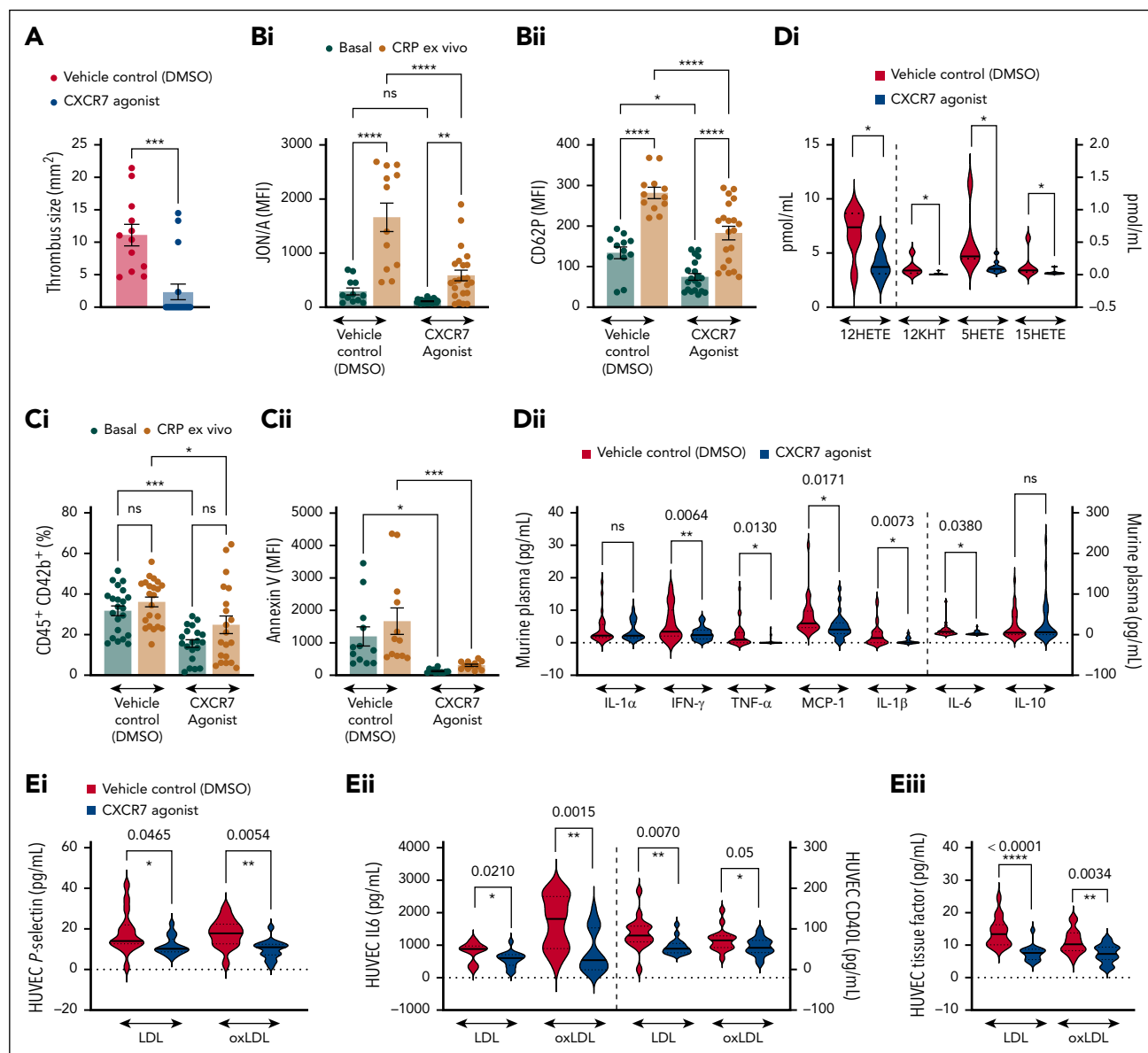


Figure 7. CXCR7 agonist modulates thrombotic and thromboinflammatory platelet responses in a murine venous thrombosis model and in endothelial cells. CXCR7 agonist (100 μ g per mouse) or vehicle control (1% DMSO) was administered IV 1 hour before the ligation of the IVC (stasis model) to induce venous thrombosis in 8- to 10-week-old wild-type C57BL/6J mice of both sexes. Blood was collected in 1:4 ACD anticoagulant to assess platelet functionality 48 hours after IVC ligation, and IVC thrombus was harvested. (A) Thrombus size in vehicle control (n = 12) and in mice administered with CXCR7 agonist (n = 17). Data are presented as mean $\pm$ standard error of the mean (SEM). *** P = .0002 by t test. (Bi) Platelet $\alpha_{IIb}\beta_{III}$ -integrin activation (JON/A-response) and (Bii) degranulation (CD62P surface expression) under basal circulating condition in blood and after ex vivo stimulation with CRP (5 μ g/mL) for 30 minutes. Measurements were performed in 1:20 diluted blood by gating on CD42b-Dylight 647-stained platelets. Data are from n = 6 vehicle control and n = 10 to 12 CXCR7 agonist-administered mice. ** P = .0054 and **** P < .0001 for (Bi) and * P = .0224 and **** P < .0001 for (Bii) by analysis of variance (ANOVA) with Šidák post hoc test. (Ci) Platelet (CD42b⁺) leukocyte (CD45⁺) aggregates estimated with anti-CD42b-FITC (platelets) and anti-CD45-APC (leukocytes) antibodies in undiluted blood under basal circulating condition and after ex vivo stimulation with CRP (5 μ g/mL) for 30 minutes in vehicle control (n = 11) and CXCR7 agonist-treated mice (n = 10). * P = .0262 and *** P = .0006 by ANOVA with Šidák post hoc test. (Cii) Phosphatidylserine-exposing annexin V⁺ procoagulant platelets under basal circulating condition and after ex vivo stimulation with CRP (5 μ g/mL) in vehicle control (n = 6) and CXCR7 agonist-treated mice (n = 6). * P = .0108 and *** P < .0005 by ANOVA with Šidák post hoc test. (B-C) Data are presented as mean $\pm$ SEM from experimental analyses performed in duplicates for each parameter tested. (Di) Levels of platelet (12-LOX), leukocyte lipoxygenase (5-LOX and 15-LOX), and platelet COX-1 pathway (12KHT)-derived oxylipins in plasma collected 48 hours after IVC ligation from vehicle control (n = 6) and CXCR7 agonist-treated mice (n = 10). * P < .05 denotes FDR-adjusted P values. (Dii) Levels of proinflammatory mediators in plasma (violin plot with median line) collected 48 hours after IVC ligation in vehicle control (n = 10) and CXCR7 agonist-treated mice (n = 10). All samples were analyzed using the Mouse Inflammation Panel LEGENDplex kit (BioLegend) in replicate measurements. Statistical significance was assessed by t test for each mediator. (Ei-Eiii) HUVECs were pretreated with CXCR7 agonist (1 μ g/mL) or vehicle control for 30 minutes, then cultured for 24 hours in the presence or absence of LDL (50 μ g/mL) or oxLDL (50 μ g/mL), and culture supernatants were collected. Data represent levels of proinflammatory mediators in the culture supernatant of HUVECs (violin plot with median line) in n = 8 experiments; all samples were analyzed using cytometric bead array using the Human Thrombosis Panel 10 LEGENDplex kit in duplicates. Statistical significance was assessed by t test for each mediator. ACD, acid-citrate-dextrose; APC, allophycocyanin; DMSO, dimethyl sulfoxide; FDR, false discovery rate; FITC, fluorescein isothiocyanate.

inhibitory effects of CXCR7 agonist on procoagulant transformation of platelets reduced FX/FXa binding; (4) while platelet-derived anticoagulant LC-CAR regulated FXa-dependent thrombin generation; (5) improved mitochondrial metabolism under the influence of CXCR7 agonist preserved $\Delta\psi_m$ and reduced mitochondrial superoxide generation which (6) consequently prevented nonenzymatic ROS-mediated lipid oxidation (LDL-to-oxLDL conversion) and peroxidation in activated platelets. Furthermore, in contrast to the effects of prothrombotic CXCL12/SDF1- α , (7) the pharmacological CXCR7 agonist regulated platelet-lipoprotein interaction and their thrombotic consequences by decreasing the surface availability of CD36 and ApoER2. (8) The ability of CXCR7 agonist to regulate thromboinflammatory and procoagulant functions, combined with its influence on anticoagulant LC-CAR, could be effective against venous thrombosis, which is substantially driven by procoagulant platelets.⁷²

Hyperlipidemia lowers the platelet activation threshold, triggering thrombotic responses,^{3,29} whereas plasma lipids are accumulated and processed intracellularly. Platelet enzymes also generate lipid mediators, signaling intermediates and agonists that perpetuate prothrombotic responses independent of stimulatory triggers from plasma lipoproteins.³ Activation-induced phospholipases generate lipid mediators, including PLC-derived DGs, PLA₂-derived atherogenic LPCs,⁷⁴ PLD-derived procoagulant LPAs,⁷⁵ and COX-1- and 12-LOX-derived oxylipins that act as platelet agonists,³ whereas the flux of FAs (arachidonic acid) through mitochondrial metabolism supports energy demands. Activated platelets also execute nonenzymatic lipid oxidation, generating oxPLs, such as oxPAPC and oxLDL,²⁹ which are increased in patients with CAD despite antiplatelet and statin therapy.²⁹ The inhibitory effects of CXCR7 agonist on nonenzymatic lipid oxidation and oxylipins are similar to those of riociguat/DEA/NO against hemin.⁷⁶ Although the increment in DGLA-derived 12-LOX metabolite 12-HETE is unique to CXCR7 agonist,¹⁴ as riociguat/DEA/NO downregulates 12-HETE.⁷⁷ Like iloprost, CXCR7 agonist engaged the platelet inhibitory IP-AC-cAMP-PKA pathway and counteracted lipoprotein-induced mitochondrial superoxide generation (Figure 1D) and ROS generation (supplemental Figure 9A) in a PKA-dependent manner, as previously reported in neutrophils.⁷⁸ This was reversed by the PKA inhibitor KT5720 (supplemental Figure 9B). Both CXCR7 agonist and iloprost prevented $\Delta\psi_m$ (Figure 1C) and intraplatelet LDL oxidation to oxLDL (Figure 1A). Therefore, enhanced platelet CXCR7 in patients with ACS could be used to interfere with both enzymatic¹⁴ and nonenzymatic lipid oxidation, limiting the generation of oxPLs with adverse effects in cardiovascular pathophysiology⁷⁹ and offering a beneficial edge over aspirin.

CXCR7 agonist counteracts platelet activatory signaling by preventing the phosphorylation of PLC γ , Src, PI3K-Akt, and p38MAPK and by inhibiting intracellular calcium mobilization downstream of CRP-GPVI.¹⁴ CXCR7 agonist administration retards arterial thrombosis, regulates thrombotic and thromboinflammatory platelet functions after MI in murine models in vivo, and inhibits platelet responses in blood from patients with ACS ex vivo without affecting plasma coagulation or tail bleeding time.¹⁴ Nevertheless, it deters procoagulant platelet-assisted thrombin generation by preventing PS externalization.¹⁴ CXCR7 agonist-imposed inhibitory effect on

procoagulant platelet activity was similar to that of iloprost, which counteracts PS exposure on platelets induced by IgGs from patients with COVID-19⁶³ and heparin-induced thrombocytopenia.⁶⁴ The present investigation demonstrated CXCR7 agonist-induced accumulation of anticoagulant LC-CARs in platelets and their subsequent release, which may also influence coagulation (Figure 3). LC-CARs (16:0, 18:0, 18:1, 18:2) affect TF- and FXa-driven clotting by interfering with the activation of prothrombin.¹⁶ Serum levels of CARs and L-carnitine are increased in patients with CAD compared to healthy subjects and are associated with CAD.⁵⁹ The CorLipid trial (NCT04580173),¹⁷ which assessed circulating CARs, demonstrated significantly lower levels of LC-CARs (16:0, 18:1, 18:2) in patients with ACS than in patients with CCS,⁵⁷ whereas patients with STEMI exhibited significantly lower levels of medium-chain CAR and LC-CARs (10:0, 16:0, 18:1, 18:2) than those with stable angina pectoris, consistent with their thrombotic disposition.⁵⁷ Short-chain CARs are elevated in patients with severe calcification and lower left ventricular ejection fraction, whereas decreased LC-CAR levels (16:0 and 18:2) are associated with high CAD complexity.¹⁷ Acetyl-C2, octanoyl-C8, and palmitoyl-C16-carnitines are significantly associated with an increased risk of cardiovascular death during 10.2 years of follow-up, whereas acetyl- and octanoyl-carnitines are associated with an increased risk of acute MI.⁵⁸ However, most of these clinical investigations have assessed serum rather than plasma CAR levels (reflecting circulatory status), part of which could have been derived from platelets during ex vivo clotting; besides, they have overlooked the anticoagulant potential of LC-CAR as a FXa inhibitor.¹⁶ Therefore, CXCR7 agonist-induced intraplatelet stockpile of LC-CARs may have clinical implications in ACS and in patients with VTE, in whom plasma levels of CARs (10:1, 12:0, 12:2, 18:1, 18:2) are reduced and remain unaltered by warfarin.¹⁶ Increasing plasma concentrations of short-chain CAR, medium-chain CAR, and LC-CAR have been reported with disease progression in patients with cardiovascular disease,⁸⁰ in contrast to the preferential enrichment of only LC-CAR induced by CXCR7 agonist. To what extent platelet-derived LC-CAR may influence coagulation and thrombotic inclination remains uncertain. Relatively increased platelet CAR levels (10:0, 12:0, 14:0, 14:1, 16:0, 16:1, and 18:2) in patients with CAD correspond to reduced platelet aggregation at baseline and incidence of major bleeding event during 3 years of follow-up.⁸¹ However, the platelet procoagulant profile is enhanced in CAD,^{4,5} which, together with lower LC-CAR levels in patients with STEMI, supports the need for a balanced regulation of platelet-assisted coagulation. Recently, reduced plasma CAR levels have been documented in patients with VITT.¹⁸ Such clinical evidence extends the relevance of anticoagulant LC-CAR across multiple thrombotic disorders. We observed a trend toward increased plasma LC-CAR levels in CXCR7 agonist-treated mice (supplemental Figure 8), which corresponded with reduced thrombus formation in the ligated IVC (Figure 7A), although lipidomics methodologies could not account for FXa-bound LC-CAR because proteins are precipitated during lipid extraction. Conversely, CXCR7 agonist administration beneficially reduced platelet/leukocyte-derived 12-, 5-, and 15-HETEs in mice (Figure 7Di); otherwise, HETE-derived oxPLs could exert procoagulant effects.¹³ CXCR7 agonist-induced platelet-derived LC-CARs may accumulate locally upon release and affect FXa-dependent thrombin

generation, as demonstrated by the counteracting effect of the CPT1 inhibitor etomoxir on CXCR7 agonist-mediated reduction of thrombin generation in PRP (Figure 3D). Although direct thrombinography data from patients with ACS and VTE are lacking because of unavoidable therapeutic use of heparin, induction of intraplatelet LC-CARs by CXCR7 agonist may replenish reduced circulating LC-CAR levels in patients with STEMI and VTE and thereby help control hypercoagulability.

Increased intraplatelet LC-CAR levels could also stem from altered mitochondrial metabolism attributed to dysregulation in carnitine shuttle enzymes (CPT1-CPT2) or inefficient β -oxidation. Higher ATP demand in thrombin-activated platelets is supplemented by phospholipase (cPLA₂)^{56,68}-generated FAs, which are converted to $\geq$ C10-CARs and metabolized through mitochondrial β -oxidation. Mitochondrial respiration and metabolomic profiling of platelets indicate that CXCR7 agonist-induced preferential accumulation of LC-CARs was not attributable to mitochondrial dysregulation. CXCR7 agonist triggered the activation of AMPK^{Thr172}-ACC^{Ser79} pathway (Figure 6D) even in resting platelets, favoring lipolysis over lipogenesis, consequently leading to depletion of lipid reserves (TG, DG, CE) and generation of LC-CAR. This, combined with reduced mitochondrial maximal respiration rate in resting platelets, resulted in the accumulation of LC-CAR, otherwise used in mitochondrial β -oxidation. The stabilizing effect of CXCR7 agonist on $\Delta\psi_m$ also prevented procoagulant PS exposure. Although the CXCR7 agonist CX771 triggers lipolysis in white adipose tissue through transcriptional regulation,⁴⁵ VUF11207 prompted the generation of free FAs and their conversion to LC-CAR by CPT1 in platelets. However, CXCR7 agonist-induced phospholipase activity and consumption of lipid reserves may eventually overwhelm mitochondrial β -oxidation, leading to the accumulation of LC-CAR and the inhibition of CPT2.⁶⁹

Noncanonical ACKR3/CXCR7⁸² ligation coordinates with G_{α_s} -coupled prostacyclin IP receptor to trigger the platelet-inhibitory AC-cAMP-PKA pathway and subsequently engages the AMPK^{Thr172}-ACC^{Ser79} axis, a regulator of mitochondrial carnitine shuttle. CXCR7 agonist-induced accumulation of LC-CARs could also result from the absence of ACC1-metabolized malonyl-CoA after pAMPK^{Ser172}-mediated inhibitory phosphorylation⁸³ of ACC^{Ser79} because malonyl-CoA prevents CAR synthesis by CPT1. The energy-sensing AMPK-ACC axis is a prime therapeutic target in metabolic syndrome used by therapeutics that activate AMPK, including metformin and thiazolidinediones.⁷¹ Inadequate activity of the AMPK-ACC axis leads to the accumulation of malonyl-CoA, which reduces FA oxidation and favors de novo lipogenesis, causing cellular and tissue accumulation of lipids.⁸⁴ Therefore, CXCR7 agonist-mediated modulation of the AMPK-ACC axis favoring lipolysis over lipogenesis appears advantageous.

In conclusion, the present investigation expands our understanding of the role of ACKR3/CXCR7 in regulating thromboinflammatory and procoagulant platelet functions through novel paradigms and presents the potential of replenishing reduced anticoagulant LC-CAR levels in patients with (immuno)thrombotic conditions. However, CXCR7 is broadly expressed in circulatory and vascular cells. Therefore, potential off-target effects on other cells and the time window of agonist

application when its availability is maximized, as in patients with ACS after MI,¹⁴ should be determined.³¹

Acknowledgments

The authors thank Markus König and Ralf Barnaba for technical assistance with the metabolomics analyses.

This project was supported by funds from the German Research Foundation DFG-374031971 (CRC-TR240), DFG-464254052 (RTG2816), DFG-335549539 (RTG2381), DFG-400989884 (NU 53/13-1), DFG-540968858; the ICePhA Graduate Program "Membrane Associated Drug Targets in Personalised Cancer Medicine," and the Robert Bosch Stiftung (Stuttgart, Germany). X.F. received financial support from the Shanghai Municipal Human Resources and Social Security Bureau (project H20250859) for collaboration with M. Lämmerhofer.

Authorship

Contribution: X.F., K.D., M. Cebo, and A.B. performed lipidomics analyses; M. Lackner performed metabolomics analyses; L.-S.M.-B. performed mitochondrial respiration analyses; T.D. and B.B. performed immunoblot analyses; J.A.B.-G. was involved in human umbilical vein endothelial cell culture; J.R., H.v.E., and H.B. performed scanning ion conductance microscopy (SICM) and immunofluorescence microscopy; S.S. performed the murine venous thrombosis model; D.R. analyzed clinical data; T.E.S. supervised SICM; M. Lämmerhofer supervised lipidomics analyses with M. Chatterjee; T.G., M.G. (CAD cohort), T.B. (procoagulant platelets), B.N., and S.B.-H. (venous thrombosis model) collaborated on the project; M.S., M.H., U.H., and E.S. supervised mitochondrial respiration and metabolomics analyses; and M. Chatterjee conceived and designed the project, supervised the study, performed experiments, analyzed data, and wrote the manuscript.

Conflict-of-interest disclosure: M.S. has received research support from Agena Bioscience, Boehringer International GmbH, and Robert Bosch GmbH and was involved in clinical trials sponsored by Gilead Sciences Inc, Corat Therapeutics GmbH, and HepaRegenix GmbH. The remaining authors declare no competing financial interests.

ORCID profiles: X.F., 0009-0007-8267-8761; A.B., 0009-0000-6107-4642; L.-S.M.-B., 0000-0001-9086-8118; B.B., 0009-0009-3666-0792; J.R., 0000-0002-1976-9245; H.B., 0000-0003-1552-8142; H.v.E., 0000-0003-0759-2056; J.A.B.-G., 0009-0002-2725-0424; B.N., 0000-0002-5995-6555; S.B.-H., 0000-0001-6049-0257; M.S., 0000-0002-9984-075X; U.H., 0000-0003-0823-9027; M.H., 0000-0001-9017-9021; T.E.S., 0000-0001-5643-8384; M. Lämmerhofer, 0000-0002-1318-0974; M. Chatterjee, 0000-0002-3428-4107.

Correspondence: Madhumita Chatterjee, Institute for Clinical and Experimental Transfusion Medicine, University Hospital Tübingen, Otfried-Müller-Str 4/1, 72076 Tübingen, Germany; email: madhumita.chatterjee@med.uni-tuebingen.de.

Footnotes

Submitted 22 July 2025; accepted 19 May 2026; prepublished online on *Blood* First Edition 15 June 2026. <https://doi.org/10.1182/blood.2025030939>.

Experimental data are available from the corresponding author, Madhumita Chatterjee (madhumita.chatterjee@med.uni-tuebingen.de), on request. Clinical data are available from Tobias Geisler (tobias.geisler@med.uni-tuebingen.de) and Dominik Rath (dominik.rath@med.uni-tuebingen.de) on request.

The online version of this article contains a data supplement.

There is a [Blood Commentary](#) on this article in this issue.

The publication costs of this article were defrayed in part by page charge payment. Therefore, and solely to indicate this fact, this article is hereby marked "advertisement" in accordance with 18 USC section 1734.

REFERENCES

- O'Donnell VB, Murphy RC, Watson SP. Platelet lipidomics: modern day perspective on lipid discovery and characterization in platelets. *Circ Res*. 2014;114(7):1185-1203.
- Chicanne G, Darcourt J, Bertrand-Michel J, Garcia C, Ribes A, Payrastre B. What can we learn from the platelet lipidome? *Platelets*. 2023;34(1):2182180.
- Chatterjee M. Platelet lipidome: dismantling the "Trojan horse" in the bloodstream. *J Thromb Haemost*. 2020;18(3):543-557.
- Pasalic L, Wing-Lun E, Lau JK, et al. Novel assay demonstrates that coronary artery disease patients have heightened procoagulant platelet response. *J Thromb Haemost*. 2018;16(6):1198-1210.
- Hajeyah AA, Protty MB, Paul D, et al. Phosphatidylthreonine is a procoagulant lipid detected in human blood and elevated in coronary artery disease. *J Lipid Res*. 2024;65(1):100484.
- Weber C, Badimon L, Mach F, van der Vorst EPC. Therapeutic strategies for atherosclerosis and atherothrombosis: past, present and future. *Thromb Haemost*. 2017;117(7):1258-1264.
- van der Meijden PEJ, Heemskerk JWM. Platelet biology and functions: new concepts and clinical perspectives. *Nat Rev Cardiol*. 2019;16(3):166-179.
- Protty MB, Jenkins PV, Collins PW, O'Donnell VB. The role of procoagulant phospholipids on the surface of circulating blood cells in thrombosis and haemostasis. *Open Biol*. 2022;12(4):210318.
- Althaus K, Marini I, Zlamal J, et al. Antibody-induced procoagulant platelets in severe COVID-19 infection. *Blood*. 2021;137(8):1061-1071.
- Ma R, Xie R, Yu C, et al. Phosphatidylserine-mediated platelet clearance by endothelium decreases platelet aggregates and procoagulant activity in sepsis. *Sci Rep*. 2017;7(1):4978.
- Uzun G, Pelzl L, Singh A, Bakchoul T. Immune-mediated platelet activation in COVID-19 and vaccine-induced immune thrombotic thrombocytopenia. *Front Immunol*. 2022;13:837629.
- Lauder SN, Allen-Redpath K, Slatter DA, et al. Networks of enzymatically oxidized membrane lipids support calcium-dependent coagulation factor binding to maintain hemostasis. *Sci Signal*. 2017;10(507):eaan2787.
- Protty MB, Tyrrell VJ, Hajeyah AA, et al. Aspirin modulates generation of procoagulant phospholipids in cardiovascular disease, by regulating LPCAT3. *J Lipid Res*. 2025;66(1):100727.
- Cebo M, Ditttrich K, Fu X, et al. Platelet ACKR3/CXCR7 favors antiplatelet lipids over an atherothrombotic lipidome and regulates thromboinflammation. *Blood*. 2022;139(11):1722-1742.
- Rohlfing AK, Kolb K, Sigle M, et al. ACKR3 regulates platelet activation and ischemia-reperfusion tissue injury. *Nat Commun*. 2022;13(1):1823.
- Deguchi H, Banerjee Y, Trauger S, et al. Acylcamitines are anticoagulants that inhibit factor Xa and are reduced in venous thrombosis, based on metabolomics data. *Blood*. 2015;126(13):1595-1600.
- Karagiannidis E, Sofidis G, Papazoglou AS, et al. Correlation of the severity of coronary artery disease with patients' metabolic profile- rationale, design and baseline patient characteristics of the CorLipid trial. *BMC Cardiovasc Disord*. 2021;21(1):79.
- Stevens H, McFadyen JD, Mellett NA, et al. Beyond platelet activation: dysregulated lipid metabolism in defining risk and pathophysiology of VITT. *Res Pract Thromb Haemost*. 2025;9(1):102677.
- Tourdot BE, Holinstat M. Targeting 12-lipoxygenase as a potential novel antiplatelet therapy. *Trends Pharmacol Sci*. 2017;38(11):1006-1015.
- Lepropre S, Kautbally S, Octave M, et al. AMPK-ACC signaling modulates platelet phospholipids and potentiates thrombus formation. *Blood*. 2018;132(11):1180-1192.
- Akkerman JW. From low-density lipoprotein to platelet activation. *Int J Biochem Cell Biol*. 2008;40(11):2374-2378.
- Obermayer G, Afonyushkin T, Binder CJ. Oxidized low-density lipoprotein in inflammation-driven thrombosis. *J Thromb Haemost*. 2018;16(3):418-428.
- Paciullo F, Momi S, Gesele P. PCSK9 in haemostasis and thrombosis: possible pleiotropic effects of PCSK9 inhibitors in cardiovascular prevention. *Thromb Haemost*. 2019;119(3):359-367.
- Pawlowska Z, Swiatkowska M, Krzeslowska J, Pawlicki L, Cierniewski CS. Increased platelet-fibrinogen interaction in patients with hypercholesterolemia and hypertriglyceridemia. *Atherosclerosis*. 1993;103(1):13-20.
- Podrez EA, Byzova TV, Febbraio M, et al. Platelet CD36 links hyperlipidemia, oxidant stress and a prothrombotic phenotype. *Nat Med*. 2007;13(9):1086-1095.
- Carvalho AC, Colman RW, Lees RS. Platelet function in hyperlipoproteinemia. *N Engl J Med*. 1974;290(8):434-438.
- Ghosh A, Murugesan G, Chen K, et al. Platelet CD36 surface expression levels affect functional responses to oxidized LDL and are associated with inheritance of specific genetic polymorphisms. *Blood*. 2011;117(23):6355-6366.
- Stellos K, Sauter R, Fahrleitner M, et al. Binding of oxidized low-density lipoprotein on circulating platelets is increased in patients with acute coronary syndromes and induces platelet adhesion to vascular wall in vivo—brief report. *Arterioscler Thromb Vasc Biol*. 2012;32(8):2017-2020.
- Chatterjee M, Rath D, Schlotterbeck J, et al. Regulation of oxidized platelet lipidome: implications for coronary artery disease. *Eur Heart J*. 2017;38(25):1993-2005.
- Harm T, Bild A, Ditttrich K, et al. Acute coronary syndrome is associated with a substantial change in the platelet lipidome. *Cardiovasc Res*. 2022;118(8):1904-1916.
- Chatterjee M. Atypical roles of the chemokine receptor ACKR3/CXCR7 in platelet pathophysiology. *Cells*. 2022;11(2):213.
- Rath D, Chatterjee M, Borst O, et al. Expression of stromal cell-derived factor-1 receptors CXCR4 and CXCR7 on circulating platelets of patients with acute coronary syndrome and association with left ventricular functional recovery. *Eur Heart J*. 2014;35(6):386-394.
- Chatterjee M, Borst O, Walker B, et al. Macrophage migration inhibitory factor limits activation-induced apoptosis of platelets via CXCR7-dependent Akt signaling. *Circ Res*. 2014;115(11):939-949.
- Rheinlaender J, Vogel S, Seifert J, et al. Imaging the elastic modulus of human platelets during thrombin-induced activation using scanning ion conductance microscopy. *Thromb Haemost*. 2015;113(2):305-311.
- Donner L, Feige T, Freiburg C, et al. Impact of amyloid-beta on platelet mitochondrial function and platelet-mediated amyloid aggregation in Alzheimer's disease. *Int J Mol Sci*. 2021;22(17):9633.
- Cebo M, Calderon Castro C, Schlotterbeck J, Gawaz M, Chatterjee M, Lammerhofer M. Untargeted UHPLC-ESI-QTOF-MS/MS analysis with targeted feature extraction at precursor and fragment level for profiling of the platelet lipidome with ex vivo thrombin-activation. *J Pharm Biomed Anal*. 2021;205:114301.
- Tsugawa H, Cajka T, Kind T, et al. MS-DIAL: data-independent MS/MS deconvolution for comprehensive metabolome analysis. *Nat Methods*. 2015;12(6):523-526.
- Paglia G, Sigurjonsson OE, Rolfsson O, et al. Comprehensive metabolomic study of platelets reveals the expression of discrete metabolic phenotypes during storage. *Transfusion*. 2014;54(11):2911-2923.
- Paglia G, Magnusdottir M, Thorlacius S, et al. Intracellular metabolite profiling of platelets: evaluation of extraction processes and chromatographic strategies. *J Chromatogr B Analyt Technol Biomed Life Sci*. 2012;898:111-120.
- Thiebold AL, Lorenz NI, Foltyn M, et al. Mammalian target of rapamycin complex 1 activation sensitizes human glioma cells to hypoxia-induced cell death. *Brain*. 2017;140(10):2623-2638.
- Hofmann U, Maier K, Niebel A, Vacun G, Reuss M, Mauch K. Identification of metabolic fluxes in hepatic cells from transient ¹³C-labeling experiments: part I.

- experimental observations. *Biotechnol Bioeng*. 2008;100(2):344-354.
42. Leuthold P, Schaeffeler E, Winter S, et al. Comprehensive metabolomic and lipidomic profiling of human kidney tissue: a platform comparison. *J Proteome Res*. 2017;16(2):933-944.
 43. Maier K, Hofmann U, Reuss M, Mauch K. Dynamics and control of the central carbon metabolism in hepatoma cells. *BMC Syst Biol*. 2010;4:54.
 44. Seifert J, von Eysmond H, Chatterjee M, Gawaz M, Schaffer TE. Effect of oxidized LDL on platelet shape, spreading, and migration investigated with deep learning platelet morphometry. *Cells*. 2021;10(11):2932.
 45. Li X, Zhu M, Penfold ME, et al. Activation of CXCR7 limits atherosclerosis and improves hyperlipidemia by increasing cholesterol uptake in adipose tissue. *Circulation*. 2014;129(11):1244-1253.
 46. Walsh TG, Harper MT, Poole AW. SDF-1alpha is a novel autocrine activator of platelets operating through its receptor CXCR4. *Cell Signal*. 2015;27(1):37-46.
 47. Leberzammer J, Agten SM, Blanchet X, et al. Targeting platelet-derived CXCL12 impedes arterial thrombosis. *Blood*. 2022;139(17):2691-2705.
 48. Chen Y, Zhang J, Cui W, Silverstein RL. CD36, a signaling receptor and fatty acid transporter that regulates immune cell metabolism and fate. *J Exp Med*. 2022;219(6):e20211314.
 49. Samovski D, Sun J, Pietka T, et al. Regulation of AMPK activation by CD36 links fatty acid uptake to beta-oxidation. *Diabetes*. 2015;64(2):353-359.
 50. Febbraio M, Hajjar DP, Silverstein RL. CD36: a class B scavenger receptor involved in angiogenesis, atherosclerosis, inflammation, and lipid metabolism. *J Clin Invest*. 2001;108(6):785-791.
 51. Bonen A, Luiken JJ, Arumugam Y, Glatz JF, Tandon NN. Acute regulation of fatty acid uptake involves the cellular redistribution of fatty acid translocase. *J Biol Chem*. 2000;275(19):14501-14508.
 52. Pohl J, Ring A, Korkmaz U, Ehehalt R, Stremmel W. FAT/CD36-mediated long-chain fatty acid uptake in adipocytes requires plasma membrane rafts. *Mol Biol Cell*. 2005;16(1):24-31.
 53. Ehehalt R, Sparla R, Kulaksiz H, Herrmann T, Fullekrug J, Stremmel W. Uptake of long chain fatty acids is regulated by dynamic interaction of FAT/CD36 with cholesterol/sphingolipid enriched microdomains (lipid rafts). *BMC Cell Biol*. 2008;9:45.
 54. Duit S, Mayer H, Blake SM, Schneider WJ, Nimpf J. Differential functions of ApoER2 and very low density lipoprotein receptor in Reelin signaling depend on differential sorting of the receptors. *J Biol Chem*. 2010;285(7):4896-4908.
 55. Cuitino L, Matute R, Retamal C, Bu G, Inestrosa NC, Marzolo MP. ApoER2 is endocytosed by a clathrin-mediated process involving the adaptor protein Dab2 independent of its Rafts' association. *Traffic*. 2005;6(9):820-838.
 56. Slatter DA, Aldrovandi M, O'Connor A, et al. Mapping the human platelet lipidome reveals cytosolic phospholipase A2 as a regulator of mitochondrial bioenergetics during activation. *Cell Metab*. 2016;23(5):930-944.
 57. Deda O, Panteris E, Meikopoulos T, et al. Correlation of serum acylcarnitines with clinical presentation and severity of coronary artery disease. *Biomolecules*. 2022;12(3):354.
 58. Strand E, Pedersen ER, Svingen GF, et al. Serum acylcarnitines and risk of cardiovascular death and acute myocardial infarction in patients with stable angina pectoris. *J Am Heart Assoc*. 2017;6(2):e003620.
 59. Gander J, Carrard J, Gallart-Ayala H, et al. Metabolic impairment in coronary artery disease: elevated serum acylcarnitines under the spotlights. *Front Cardiovasc Med*. 2021;8:792350.
 60. Campbell SE, Tandon NN, Woldegiorgis G, Luiken JJ, Glatz JF, Bonen A. A novel function for fatty acid translocase (FAT)/CD36: involvement in long chain fatty acid transfer into the mitochondria. *J Biol Chem*. 2004;279(35):36235-36241.
 61. Bezaire V, Bruce CR, Heigenhauser GJ, et al. Identification of fatty acid translocase on human skeletal muscle mitochondrial membranes: essential role in fatty acid oxidation. *Am J Physiol Endocrinol Metab*. 2006;290(3):E509-E515.
 62. Zlamal J, Aliotta A, Alberio L, Chen V, Bakchoul T; SSC Platelet Immunology of the ISTH. Diagnostic value of antibody-induced procoagulant platelets in heparin-induced thrombocytopenia: communication from the ISTH SSC Subcommittee on Platelet Immunology. *J Thromb Haemost*. 2024;22(3):860-868.
 63. Zlamal J, Althaus K, Jaffal H, et al. Upregulation of cAMP prevents antibody-mediated thrombus formation in COVID-19. *Blood Adv*. 2022;6(1):248-258.
 64. Zlamal J, Singh A, Weich K, et al. Platelet phosphatidylserine is the critical mediator of thrombosis in heparin-induced thrombocytopenia. *Haematologica*. 2023;108(10):2690-2702.
 65. Chatterjee M, Seizer P, Borst O, et al. SDF-1alpha induces differential trafficking of CXCR4-CXCR7 involving cyclophilin A, CXCR7 ubiquitination and promotes platelet survival. *FASEB J*. 2014;28(7):2864-2878.
 66. Aibibula M, Naseem KM, Sturmey RG. Glucose metabolism and metabolic flexibility in blood platelets. *J Thromb Haemost*. 2018;16(11):2300-2314.
 67. Chapman AG, Atkinson DE. Stabilization of adenylate energy charge by the adenylate deaminase reaction. *J Biol Chem*. 1973;248(23):8309-8312.
 68. Ravi S, Chacko B, Sawada H, et al. Metabolic plasticity in resting and thrombin activated platelets. *PLoS One*. 2015;10(4):e0123597.
 69. Fan X, Wang Y, Cai X, et al. CPT2 K79 acetylation regulates platelet life span. *Blood Adv*. 2022;6(17):4924-4935.
 70. Oakhill JS, Steel R, Chen ZP, et al. AMPK is a direct adenylate charge-regulated protein kinase. *Science*. 2011;332(6036):1433-1435.
 71. Hardie DG, Ross FA, Hawley SA. AMPK: a nutrient and energy sensor that maintains energy homeostasis. *Nat Rev Mol Cell Biol*. 2012;13(4):251-262.
 72. Kaiser R, Dewender R, Mulkers M, et al. Procoagulant platelet activation promotes venous thrombosis. *Blood*. 2024;144(24):2546-2553.
 73. Poredos P, Jezovnik MK. Dyslipidemia, statins, and venous thromboembolism. *Semin Thromb Hemost*. 2011;37(8):897-902.
 74. Diehl P, Nienaber F, Zaldivia MTK, et al. Lysophosphatidylcholine is a major component of platelet microvesicles promoting platelet activation and reporting atherosclerotic plaque instability. *Thromb Haemost*. 2019;119(8):1295-1310.
 75. Bolen AL, Naren AP, Yarlaga S, et al. The phospholipase A1 activity of lysophospholipase A-I links platelet activation to LPA production during blood coagulation. *J Lipid Res*. 2011;52(5):958-970.
 76. Rohlfing AK, Kremser M, Schaale D, et al. cGMP modulates hemin-mediated platelet death. *Thromb Res*. 2024;234:63-74.
 77. Fu X, Knappe C, Rohlfing AK, Gawaz MP, Lammerhofer M. Non-enantioselective, enantioselective, and two-dimensional liquid chromatography coupled with tandem mass spectrometry for the study of stereochemical disposition of oxylipins in cGMP-regulated hemin-treated platelets. *J Pharm Biomed Anal*. 2024;248:116328.
 78. Raad H, Mouawia H, Hassan H, et al. The protein kinase A negatively regulates reactive oxygen species production by phosphorylating gp91phox/NOX2 in human neutrophils. *Free Radic Biol Med*. 2020;160:19-27.
 79. Salomon RG. Structural identification and cardiovascular activities of oxidized phospholipids. *Circ Res*. 2012;111(7):930-946.
 80. Moskaleva NE, Shestakova KM, Kukharensko AV, et al. Target metabolome profiling-based machine learning as a diagnostic approach for cardiovascular diseases in adults. *Metabolites*. 2022;12(12):1185.
 81. Harm T, Dittich K, Brun A, et al. Large-scale lipidomics profiling reveals characteristic lipid signatures associated with an increased

- cardiovascular risk. *Clin Res Cardiol.* 2023; 112(11):1664-1678.
82. Nurnberg B, Beer-Hammer S, Reisinger E, Leiss V. Non-canonical G protein signaling. *Pharmacol Ther.* 2024;255:108589.
83. Aslam M, Ladilov Y. Emerging role of cAMP/ AMPK signaling. *Cells.* 2022;11(2):308.
84. Zordoky BN, Nagendran J, Pulinilkunnil T, et al. AMPK-dependent inhibitory phosphorylation of ACC is not essential for maintaining myocardial fatty acid oxidation. *Circ Res.* 2014;115(5):518-524.

© 2026 American Society of Hematology. Published by Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.